OceanStor Dorado 6.1.6 MIB Interface

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1 Overview

This document describes the management information bases (MIBs) of Huawei storage devices, covering the MIBs' functions, nodes, tables, and alarms.

1.1 Intended Audience

1.2 Reading Guide

1.1 Intended Audience

This document is intended for:

- Application program development personnel

1.2 Reading Guide

This document gives the MIB overview and multiple MIB reference documents:

Chapter	Content
1. MIB Introduction	Network management, SNMP-based network management, Simple Network Management Protocol (SNMP), and MIB concepts and operations.
2. MIB References	MIB-based naming.

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2 MIB Introduction

2.1 Network Management
2.2 SNMP-based Network Management
2.3 SNMP Introduction
2.4 MIB Introduction
2.5 MIB Browser Introduction
2.6 Query Return Value Description

2.1 Network Management

The ever expanding network and increasing network devices present a major challenge in managing networks effectively and providing high-quality network services. Network management is vital to the proper functionality of a network solution.

Network management has four elements:

- Managed node: The device that is to be managed on the network.
- Agent: The software or hardware that tracks managed nodes.
- Manager: Communicates with the Agents in different managed nodes and displays the status of the managed nodes.
- Network management protocol: Enables information exchange between the Manager and the Agents.

Currently, the most widely adopted network management protocol for TCP/IP network is SNMP.

2.2 SNMP-based Network Management

The SNMP-based network management involves four parts:

- Network Management Station (NMS)

  The NMS is an independent device that runs network management programs. The network management programs provide at least one man-machine

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interface for network administrators to perform network management operations.

- SNMP Agent

  The SNMP Agent is a software module that resides on a managed device. The Agent receives and processes the packets sent by the NMS, and then sends response packets to the NMS. Under emergent conditions, the Agent sends **trap** packets proactively to communicate with the NMS.

- SNMP

  The SNMP works at the application layer of the TCP/IP network. It enables information exchange between the NMS and managed devices.

- MIB

  The MIB is a collection of managed objects. It is a bridge for communication between the NMS and the Agent. In addition, it enables standard interconnection between network management software and managed devices. Each Agent has a MIB whose object values can be abstracted or set by the NMS.

The previous four parts communicate with one another as follows:

- The NMS uses SNMP to communicate with the Agent and read or modify the MIB. In this way, the NMS monitors and manages the network devices.
- SNMP enables communication between the NMS and the Agent. The Protocol Data Unit (PDU) of SNMP implements information exchange. The SNMP does not handle data transfer by itself, but User Datagram Protocol (UDP) does instead.
- The Agent communicates with the NMS to maintain the MIB, to monitor and manage each module of the device.
- The MIB stores information about each module of the device. The NMS accesses the information to monitor and manage the device.

2.3 SNMP Introduction

2.3.1 Versions

SNMP has three versions: SNMPv1, SNMPv2c, and SNMPv3.

SNMPv1 and SNMPv2c use community-based SNMP authentication. The NMS uses the community name list to control the access to devices. The Agent does not check whether the sender uses an authorized community name. In addition, SNMP messages are transmitted without encryption. As a result, SNMPv1 and SNMPv2c cannot secure authentication or privacy.

SNMPv2c is an enhanced version of SNMPv1. Compared with SNMPv1, SNMPv2c supports more operations, data types, communications protocols, and error handling codes.

SNMPv3 provides all functions of SNMPv1 and SNMPv2c, and offers a more advanced security mechanism for authentication and encryption.

SNMPv3 ensures data security and access control.

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SNMPv3 enables message-level data security in terms of the three aspects:

- Data integrity

  Data modification must be authorized beforehand. Data sequence modification must be within the specific scope.
- Data origin authentication

  This authentication makes clear which user the received data comes from. SNMPv3 uses user-based security authentication. It verifies a user who sends messages, but not an application program that generates messages.
- Data verification check

  The NMS or Agent checks the time a received message was generated. The message will not be accepted if the difference between the message time and the system time exceeds the specified range. Data verification check protects in-transfer messages against being tampered with, and prevents the reception and handling of malicious messages.

  SNMPv3 implements access control based on protocol operations, and controls the access to managed objects.

2.3.2 Packet Handling

The Agent receives request packets from the NMS through UDP port 161.

The Agent processes a received packet as follows:

1. Decodes the packet. According to ASN.1 encoding rules, the Agent generates a packet that is represented in the internal data structure. If decoding fails due to an expected error, the Agent discards the packet.
2. Checks the SNMP version. The Agent compares the version of the received packet with the list of supported versions. If the version is excluded from the list, the Agent discards the packet.
3. Authenticates the community name. The Agent obtains the community name from the packet. The community name is filled by the NMS sending this packet request. If the community name is excluded from the list of supported community names, the Agent discards the packet and generates a **trap** packet.
4. Abstracts the PDU. The Agent abstracts the PDU from the successfully authenticated ASN.1 object. If the abstraction fails, the Agent discards the packet.
5. Handles the PDU. The Agent handles the PDU according to its SNMP entity to learn the MIB tree node of the managed **variable**. Then the Agent obtains the value of the managed **variable** from the corresponding module to generate a response packet. After that, the Agent encodes the response packet and sends it to the NMS.
6. The NMS receives the response packet and handles it in a similar way. Then the NMS displays a result.

2.4 MIB Introduction

The MIB is a collection of managed objects. It defines attributes of managed objects. The attributes are:

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- Name of the object
- Permission for the object
- Data type of the object

The Structure of Management Information (SMI) specifies the definition and organization of each managed object. In addition, it specifies the types of data applicable to the MIB, such as **Counter** and **Gauge**.

The MIB designates network element variables, specifically, information that can be queried and configured by the NMS. The MIB gives a data structure for the collection of possible managed objects.

2.5 MIB Browser Introduction

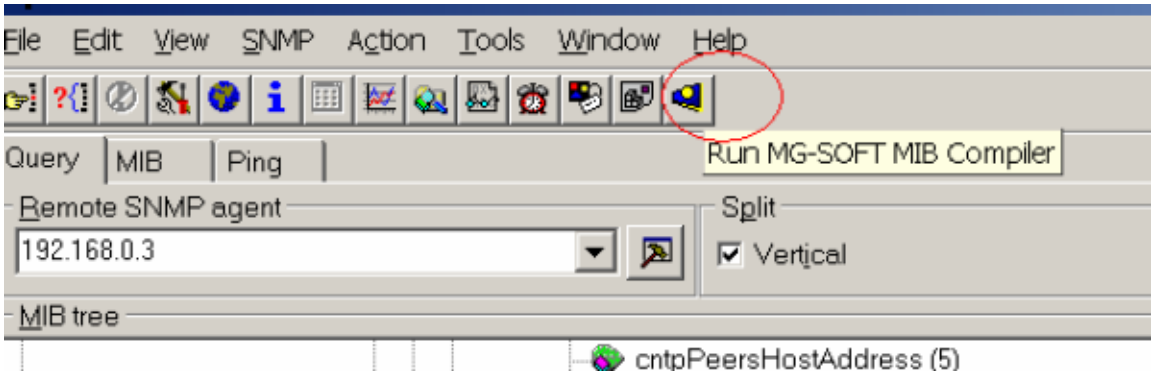
When you load MIB files, load the ones to be imported first.

Here we use the MG-SOFT MIB Browser Professional SNMPv3 Edition as an example to explain how to load MIB files.

2.5.1 Compiling and Loading MIB Files

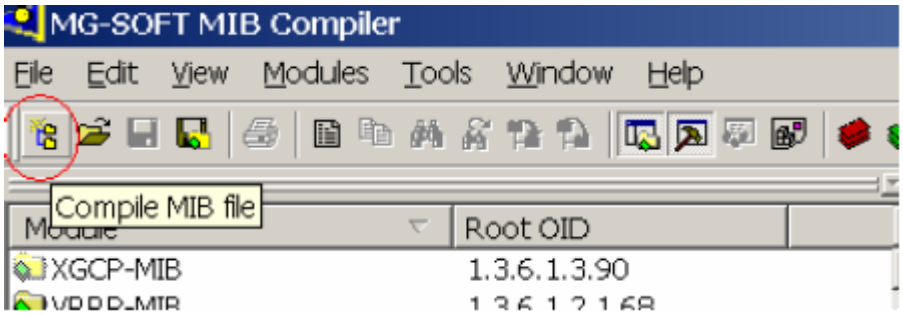
Step 1 Start the MG-SOFT MIB browser. On the **MIB Browser** window, click the **Run MG-SOFT MIB Compiler** icon, as shown in **Figure 2-1**:

Figure 2-1 MIB browser window



Step 2 On the **MG-SOFT MIB Compiler** window that is displayed, click the **Compile MIB file** icon, as shown in **Figure 2-2**:

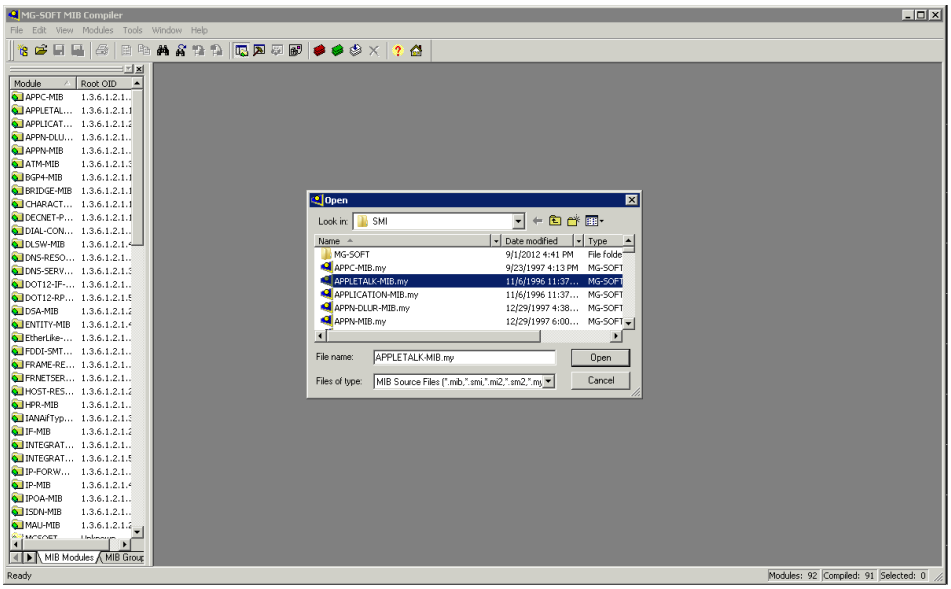
Figure 2-2 Selecting MIB compiling



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Step 3 Select the MIB file that you want to compile.

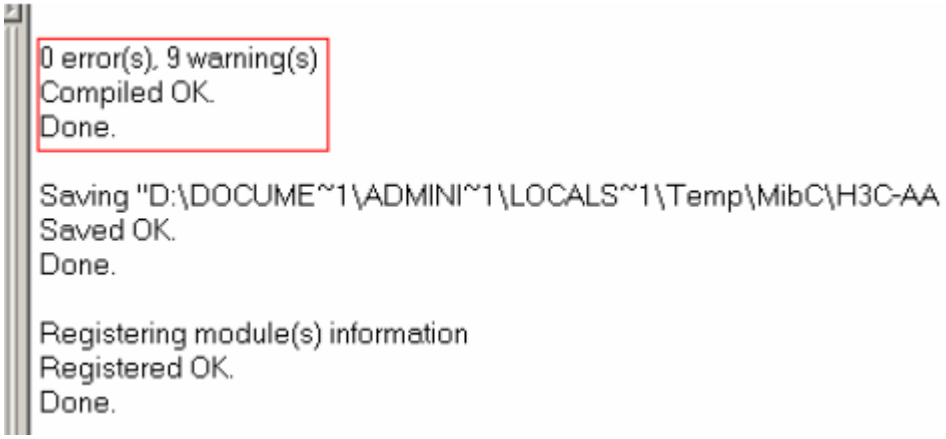
Figure 2-3 Selecting a MIB file



Step 4 Compile the selected MIB file.

Select the MIB file that you want to compile and click **Open**. If the compile succeeds, the following information is displayed:

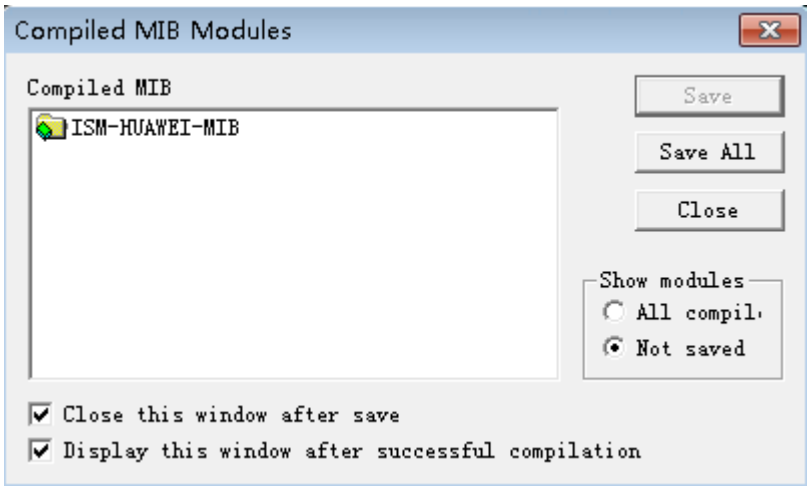
Figure 2-4 Information indicating a successful compile



Step 5 Save the successfully compiled MIB file. The MIB file will be saved in the default folder.

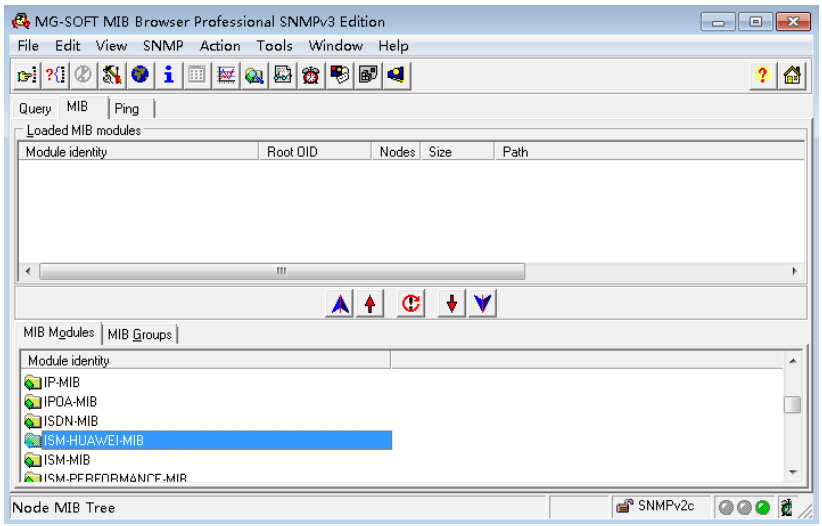
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Figure 2-5 Saving the MIB file



Step 6 Select the MIB file **ISM-HUAWEI-MIB** in **MIB/MIB Modules** column, click the red arrow to upload MIB file into the MIB Browser.

----End



2.5.2 Configuring SNMP protocol parameters

Timeout: Indicates the timeout interval of request packets. The value ranges from 1s to 30s. 15s is recommended.

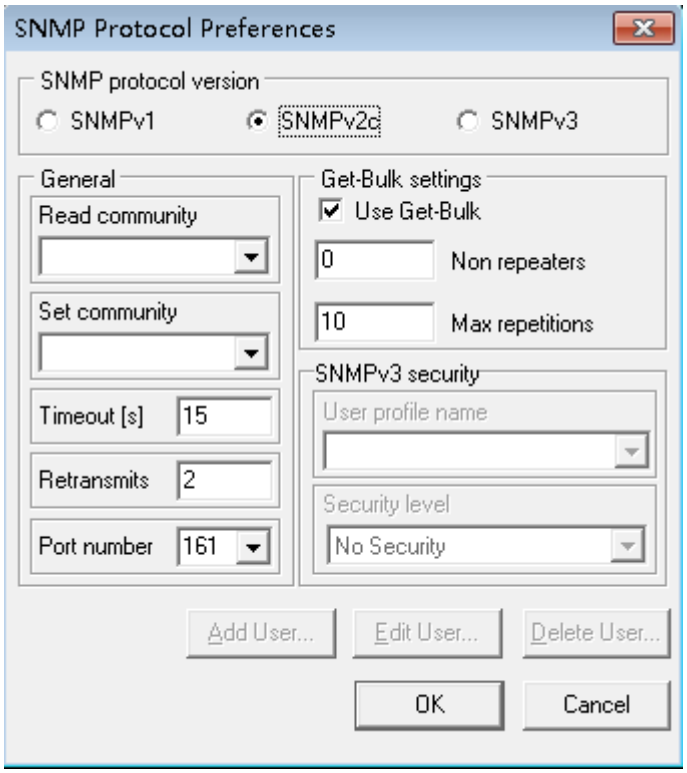
Retransmits: Indicates the resending times of request packets after the request packets timed out. The value ranges from 0 to 10. 2 is recommended. You can set the value based on the network environment.

Non repeaters: Indicates that the Get Next operation is executed only once for the first **Non repeaters** parameters in binding parameters. If the parameter value is greater than the total number of binding parameters, the total number of binding parameters prevails. If the parameter value is smaller than 0, **0** prevails. You are advised to set the parameter value to **0**. Parameter settings vary depending on site requirements.

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Max repetitions: Indicates the maximum number of repeated Get Next operations performed for all parameters except the first **Non repeaters** parameters in binding parameters. The value ranges from 1 to 50. 10 is recommended. A too large value results in **tooBig** errors.

Note: Parameters **Non repeaters** and **Max repetitions** are unavailable for the SNMPv1 protocol.



2.6 Query Return Value Description

There are two forms of snmp query return value:

One is OID + index value + data:

```
33: hwStorageFCPortID.1122304 (Unsigned32) 1122304
34: hwStorageFCPortID.1122305 (Unsigned32) 1122305
35: hwStorageFCPortID.1122306 (Unsigned32) 1122306
36: hwStorageFCPortID.1122307 (Unsigned32) 1122307
```

The other is OID + index value length + index value + data:

```
43: hwPerfDiskIndex.1.48 (OCTET STRING) 0 [
44: hwPerfDiskIndex.1.49 (OCTET STRING) 1 [
45: hwPerfDiskIndex.1.50 (OCTET STRING) 2 [
46: hwPerfDiskIndex.1.51 (OCTET STRING) 3 [
```

The index value is generated internally by us as an index of data for querying. For example, when querying hard disk data, there are multiple hard disks in the system. The internal query will be OID + index value query to avoid data confusion on multiple disks.

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The index value is the value obtained by the first OID in each table. The type of the index value is the cause of the difference in the return value.

1. The type of the index value is a string, then the presentation form is: ascll code length + ascll code value of the index value (because the type is a string, its length is required to continue the query).

2. The type of the index value is non-string, so the presentation form is: index value (non-string, the index value can continue to query)

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3 RFC MIB Support

3.1 RFC 1213 mib-2 System Module

3.1 RFC 1213 mib-2 System Module

sysDescr

Node	Data Type	Meaning	Permission	Status
sysDescr OID: 1.3.6.1.2.1.1.1	OCTET STRING	System description	Read-only	Current

sysObjectID

Node	Data Type	Meaning	Permission	Status
sysObjectID OID: 1.3.6.1.2.1.1.2	Object Identifier	System OID	Read-only	Current

sysUpTime

Node	Data Type	Meaning	Permission	Status
sysUpTime OID: 1.3.6.1.2.1.1.3	Timeticks	Running time of the SNMP service. Unit: 1% second.	Read-only	Current

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sysName

Node	Data Type	Meaning	Permission	Status
sysName OID: 1.3.6.1.2.1.1.5	OCTET STRING	System name	Read-write	Current

sysLocation

Node	Data Type	Meaning	Permission	Status
sysLocation OID: 1.3.6.1.2.1.1.6	OCTET STRING	System location	Read-write	Current

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4 MIB References for Huawei Nodes (Standard Enterprise Node: 2011)

4.1 Common Module
4.2 Alarm Module
4.3 Alarm Module(alarm:251)
4.4 Storage Hardware Modules
4.5 TOPO Module
4.6 SAN Module

4.1 Common Module

patch version

Node	Data Type	Meaning	Permission	Status
patchversion OID: 1.3.6.1.4.1.2011.2.251.1.1.7	OCTET STRING	Patch version.	Read-only	Current

4.2 Alarm Module

hwIsmActiveAlarmInfoTable

This table saves active alarms, which can be queried by the NMS.

The indexes to this table are **1: hwIsmActiveAlarmInfoNodeCode** and **2: hwIsmActiveAlarmInfoSerialNo**.

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The OID of this table is 1.3.6.1.4.1.2011.2.91.10.1.1.

Node	Data Type	Meaning	Permission	Status
hwIsmActiveAlarmInfoNodeCode OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.1	OCTET STRING	ID of the node where an alarm occurs.	Read-only	Current
hwIsmActiveAlarmInfoLocationInfo OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	Read-only	Current
hwIsmActiveAlarmInfoRestoreAdvice OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.3	OCTET STRING	Alarm handling suggestion.	Read-only	Current
hwIsmActiveAlarmInfoTitle OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.4	OCTET STRING	Alarm information title.	Read-only	Current
hwIsmActiveAlarmInfoType OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.5	Integer	Alarm type: 2-EquipmentAlarm	Read-only	Current
hwIsmActiveAlarmInfoLevel OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.6	Integer	Alarm severity of I2000 network management software: 1. -criticalAlarm 2. -majorAlarm 3. -minorAlarm 4. -warningAlarm	Read-only	Current
hwIsmActiveAlarmInfoAlarmID OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.7	Gauge	Alarm ID of I2000 network management software.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwIsmActiveAlarmInfoOccurTime OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.8	OCTET STRING	Time when an alarm was generated.	Read-only	Current
hwIsmActiveAlarmInfoSerialNo OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.9	Gauge	Alarm serial number (SN)	Read-only	Current
hwIsmActiveAlarmInfoAddtionInfo OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.10	OCTET STRING	Alarm cause	Read-only	Current
hwIsmActiveAlarmInfoCategory OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.11	Integer	Alarm category: 1. -faultAlarm 2. -resumeAlarm 3. -eventAlarm	Read-only	Current
hwIsmActiveAlarmInfoLocalAlarmID OID: 1.3.6.1.4.1.2011.2.91.10.1.1.1.12	Counter64	Original alarm ID	Read-only	Current

hwIsmTrapTargetAddrTable

This table gives information about alarm forwarding addresses, involving IP addresses, ports, and versions.

The index to this table is **hwIsmTrapTargetAddrIndex**.

The OID of this table is 1.3.6.1.4.1.2011.2.91.10.4.1.

Node	Data Type	Meaning	Permission	Status
hwIsmTrapTargetAddrIPAddr OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.1	IpAddress	Destination address for trap forwarding. Only support IPv4 format.	Read-create	Current

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hwIsmTrapTargetAddrPort OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.2	Integer	Port of the destination address for trap forwarding. The value can be 1 to 65535.	Read-create	Current
hwIsmTrapTargetAddrRowStatus OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.3	Integer	Row status. The value can be: ● 1: Modify the Trap forwarding destination address ● 4: Add the Trap forwarding destination address ● 6: Delete the Trap forwarding destination address	Read-create	Current
hwIsmTrapTargetAddrIndex OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.4	OCTET STRING	Index of the destination address for trap forwarding. The value can be 0 to 3.	Read-create	Current
hwIsmTrapTargetAddrTrapVer OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.5	Integer	Version of the destination address for trap forwarding. The value can be: ● 1: SNMPv1 ● 2: SNMPv2c ● 3: SNMPv3 When configuring the destination address for trap forwarding, if this field is not filled, the default is SNMPv2c.	Read-create	Current
hwIsmTrapTargetAddrIPAddrNew OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.6	OCTET STRING	New destination address for trap forwarding. Support IPv4 or IPv6 format. If **hwIsmTrapTargetAddrIPAddr** and **hwIsmTrapTargetAddrIPAddrNew** fields are filled with values, take the value of the **hwIsmTrapTargetAddrIPAddrNew** field by default.	Read-create	Current

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hwIsmTrapTargetAddr-TrapType OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.7	Integer	Type of the destination address for trap forwarding. The value can be: ● 1: Parsed. Sends parsed alarms to the destination address for trap forwarding (Trap OID:1.3.6.1.4.1.2011.2.91.10.2.1.0.1) ● 2: Original. Sends original alarms to the destination address for trap forwarding (Trap OID:1.3.6.1.4.1.34774.4.1.4.2). ● 3: All. sends Parsed, Original, Parsed_time_string and Original_time_string alarms to the destination address for trap forwarding. ● 5: Parsed_time_string. Sends parsed alarms to the destination address for trap forwarding (Trap OID: 1.3.6.1.4.1.2011.2.251.20.1.2.1) ● 6: Original_time_string. Sends original alarms to the destination address for trap forwarding (Trap OID:1.3.6.1.4.1.2011.2.251.20.2.1.1). When configuring the destination address for trap forwarding, if this field is not filled, the default is Parsed.	Read-create	Current

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hwIsmTrapTargetAddrUSMUserName OID: 1.3.6.1.4.1.2011.2.91.10.4.1.1.8	OCTET STRING	USM user name of the destination address for trap forwarding. When the trap forwarding address type is SNMPv3, this field is mandatory and users must ensure a specified USM user exists in the system. Otherwise, this operation fails to be performed. If this field is mandatory, and the parameter is not delivered or no value is specified for the parameter, the system default value is **Kaimse**.	Read-create	Current

hwIsmClearedAlarmConfirm

Node	Data Type	Meaning	Permission	Status
hwIsmClearedAlarmConfirm OID: 1.3.6.1.4.1.2011.2.91.10.1.2	Gauge	Confirmation of an active alarm. **Set**: Complete the SN of an active alarm. If the SN exists, clear this alarm and a clear alarm is reported. If the SN does not exist, an error message is returned. **Get**: Returns a fixed value: 0, the value is meaningless.	Read-write	Current

hwIsmAlarmReporting

The node is used for reporting the parsed alarm.

The OID of this node is 1.3.6.1.4.1.2011.2.91.10.2.1.0.1.

Bound Variable	Data Type	Meaning	Permission	Status
hwIsmReportingAlarmNodeCode OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.1	OCTET STRING	ID of the node where an alarm occurs.	accessible-for-notify	current

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hwIsmReportingAlarmLocationInfo OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	accessible-for-notify	current
hwIsmReportingAlarmRestoreAdvice OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.3	OCTET STRING	Alarm handling suggestion.	accessible-for-notify	current
hwIsmReportingAlarmFaultTitle OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.4	OCTET STRING	Alarm information title.	accessible-for-notify	current
hwIsmReportingAlarmFaultType OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.5	Integer32	Alarm type: 2-EquipmentAlarm 3-EventAlarm	accessible-for-notify	current
hwIsmReportingAlarmFaultLevel OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.6	Integer32	Alarm severity of I2000 network management software: 1-criticalAlarm 2-majorAlarm 3-minorAlarm 4-warningAlarm 5-eventAlarm	accessible-for-notify	current
hwIsmReportingAlarmAlarmID OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.7	Gauge	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwIsmReportingAlarmFaultTime OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.8	DateAndTime	Time when an alarm was generated.	accessible-for-notify	current
hwIsmReportingAlarmSerialNo OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.9	Gauge	Alarm serial number (SN)	accessible-for-notify	current
hwIsmReportingAlarmAdditionInfo OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.10	OCTET STRING	Alarm cause	accessible-for-notify	current

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hwIsmReportingAlarmFaultCategory OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.11	Integer32	Alarm category: 1-faultAlarm 2-resumeAlarm 3-eventAlarm	accessible-for-notify	current
hwIsmReportingAlarmLocationAlarmID OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.12	Counter64	Original alarm ID	accessible-for-notify	current
hwIsmReportingAlarmProductModel OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.13	Integer32	Product Model	accessible-for-notify	current
hwIsmReportingAlarmProductSN OID:1.3.6.1.4.1.2011.2.91.10.3.1.1.14	OCTET STRING	SN	accessible-for-notify	current

4.3 Alarm Module(alarm:251)

hwStorageActiveAlarmInfoTable

This table saves active alarms, which can be queried by the NMS.

The indexes to this table are **1: hwStorageActiveAlarmInfoNodeCode** and **2: hwStorageActiveAlarmInfoSerialNo**.

The OID of this table is 1.3.6.1.4.1.2011.2.251.20.1.1.1.

Node	Data Type	Meaning	Permission	Status
hwStorageActiveAlarmInfoNodeCode OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.1.1	OCTET STRING	ID of the node where an alarm occurs.	Read-only	Current
hwStorageActiveAlarmInfoLocationInfo OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.1.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageActiveAlarmInfoRestoreAdvice OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.3	OCTET STRING	Alarm handling suggestion.	Read-only	Current
hwStorageActiveAlarmInfoTitle OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.4	OCTET STRING	Alarm information title.	Read-only	Current
hwStorageActiveAlarmInfoType OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.5	Integer	Alarm type: 2-EquipmentAlarm	Read-only	Current
hwStorageActiveAlarmInfoLevel OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.6	Integer	Alarm severity of I2000 network management software: 1. -criticalAlarm 2. -majorAlarm 3. -minorAlarm 4. -warningAlarm	Read-only	Current
hwStorageActiveAlarmInfoAlarmID OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.7	Gauge	Alarm ID of I2000 network management software.	Read-only	Current
hwStorageActiveAlarmInfoOccurTime OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.8	OCTET STRING	Time when an alarm was generated.	Read-only	Current
hwStorageActiveAlarmInfoSerialNo OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.9	Gauge	Alarm serial number (SN)	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageActiveAlarmInfoAddtionInfo OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.10	OCTET STRING	Alarm cause	Read-only	Current
hwStorageActiveAlarmInfoCategory OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.11	Integer	Alarm category: 1. -faultAlarm 2. -resumeAlarm 3. -eventAlarm	Read-only	Current
hwStorageActiveAlarmInfoLocalAlarmID OID: 1.3.6.1.4.1.2011.2.251.20.1.1.1.12	Counter64	Original alarm ID	Read-only	Current

hwStorageAlarmReporting

The node is used for reporting the parsed alarm.

The OID of this node is 1.3.6.1.4.1.2011.2.251.20.1.2.1.

Bound Variable	Data Type	Meaning	Permission	Status
hwStorageReportingAlarmNodeCode OID:1.3.6.1.4.1.2011.2.251.20.1.3.1	OCTET STRING	ID of the node where an alarm occurs.	accessible-for-notify	current
hwStorageReportingAlarmLocationInfo OID:1.3.6.1.4.1.2011.2.251.20.1.3.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	accessible-for-notify	current
hwStorageReportingAlarmRestoreAdvice OID:1.3.6.1.4.1.2011.2.251.20.1.3.3	OCTET STRING	Alarm handling suggestion.	accessible-for-notify	current
hwStorageReportingAlarmFaultTitle OID:1.3.6.1.4.1.2011.2.251.20.1.3.4	OCTET STRING	Alarm information title.	accessible-for-notify	current

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hwStorageReportingAlarmFaultType OID:1.3.6.1.4.1.2011.2.251.20.1.3.5	Integer32	Alarm type: 2-EquipmentAlarm 3-EventAlarm	accessible-for-notify	current
hwStorageReportingAlarmFaultLevel OID:1.3.6.1.4.1.2011.2.251.20.1.3.6	Integer32	Alarm severity of I2000 network management software: 1-criticalAlarm 2-majorAlarm 3-minorAlarm 4-warningAlarm 5-eventAlarm	accessible-for-notify	current
hwStorageReportingAlarmAlarmID OID:1.3.6.1.4.1.2011.2.251.20.1.3.7	Gauge	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwStorageReportingAlarmFaultTime OID:1.3.6.1.4.1.2011.2.251.20.1.3.8	OCTET STRING	Time when an alarm was generated.	accessible-for-notify	current
hwStorageReportingAlarmSerialNo OID:1.3.6.1.4.1.2011.2.251.20.1.3.9	Gauge	Alarm serial number (SN)	accessible-for-notify	current
hwStorageReportingAlarmAdditionInfo OID:1.3.6.1.4.1.2011.2.251.20.1.3.10	OCTET STRING	Alarm cause	accessible-for-notify	current
hwStorageReportingAlarmFaultCategory OID:1.3.6.1.4.1.2011.2.251.20.1.3.11	Integer32	Alarm category: 1-faultAlarm 2-resumeAlarm 3-eventAlarm	accessible-for-notify	current
hwStorageReportingAlarmLocationAlarmID OID:1.3.6.1.4.1.2011.2.251.20.1.3.12	Counter64	Original alarm ID	accessible-for-notify	current
hwStorageReportingAlarmProductModel OID:1.3.6.1.4.1.2011.2.251.20.1.3.13	Integer32	Product Model	accessible-for-notify	current

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hwStorageReportingAlarm-ProductSN OID:1.3.6.1.4.1.2011.2.251.20.1.3.14	OCTET STRING	SN	accessible-for-notify	current

hwStorageAlarmReportingPrecise

The node is used for reporting the parsed alarms. The alarm ID of each alarm corresponds to an independent OID, and different OIDs are used for alarm generation and clearance.

For details about the OID of the node, see the *Alarm OIDList (Only for Parsed_time_string_precise).xlsx*.

Bound Variable	Data Type	Meaning	Permission	Status
hwStorageReportingAlarm-NodeCode OID:1.3.6.1.4.1.2011.2.251.20.1.3.1	OCTET STRING	ID of the node where an alarm occurs.	accessible-for-notify	current
hwStorageReportingAlarm-LocationInfo OID:1.3.6.1.4.1.2011.2.251.20.1.3.2	OCTET STRING	Alarm location information. The format is: Name1 = Value1, …, NameN = ValueN	accessible-for-notify	current
hwStorageReportingAlarm-RestoreAdvice OID:1.3.6.1.4.1.2011.2.251.20.1.3.3	OCTET STRING	Alarm handling suggestion.	accessible-for-notify	current
hwStorageReportingAlarm-FaultTitle OID:1.3.6.1.4.1.2011.2.251.20.1.3.4	OCTET STRING	Alarm information title.	accessible-for-notify	current
hwStorageReportingAlarm-FaultType OID:1.3.6.1.4.1.2011.2.251.20.1.3.5	Integer32	Alarm type: 2-EquipmentAlarm 3-EventAlarm	accessible-for-notify	current

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hwStorageReportingAlarmFaultLevel OID:1.3.6.1.4.1.2011.2.251.20.1.3.6	Integer32	Alarm severity of I2000 network management software: 1-criticalAlarm 2-majorAlarm 3-minorAlarm 4-warningAlarm 5-eventAlarm	accessible-for-notify	current
hwStorageReportingAlarmAlarmID OID:1.3.6.1.4.1.2011.2.251.20.1.3.7	Gauge	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwStorageReportingAlarmFaultTime OID:1.3.6.1.4.1.2011.2.251.20.1.3.8	OCTET STRING	Time when an alarm was generated.	accessible-for-notify	current
hwStorageReportingAlarmSerialNo OID:1.3.6.1.4.1.2011.2.251.20.1.3.9	Gauge	Alarm serial number (SN)	accessible-for-notify	current
hwStorageReportingAlarmAdditionInfo OID:1.3.6.1.4.1.2011.2.251.20.1.3.10	OCTET STRING	Alarm cause	accessible-for-notify	current
hwStorageReportingAlarmFaultCategory OID:1.3.6.1.4.1.2011.2.251.20.1.3.11	Integer32	Alarm category: 1-faultAlarm 2-resumeAlarm 3-eventAlarm	accessible-for-notify	current
hwStorageReportingAlarmLocationAlarmID OID:1.3.6.1.4.1.2011.2.251.20.1.3.12	Counter64	Original alarm ID	accessible-for-notify	current
hwStorageReportingAlarmProductModel OID:1.3.6.1.4.1.2011.2.251.20.1.3.13	Integer32	Product Model	accessible-for-notify	current
hwStorageReportingAlarmProductSN OID:1.3.6.1.4.1.2011.2.251.20.1.3.14	OCTET STRING	SN	accessible-for-notify	current

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hwStorageEventType

This table is used exclusively for ISM management software and is not available to others.

The OID of this node is 1.3.6.1.4.1.2011.2.251.20.2.1.1.

Bound Variable	Data Type	Meaning	Permission	Status
hwStorageTrapEvent-Type OID:1.3.6.1.4.1.2011.2.251.20.2.2.1	Gauge	Alarm type. The value can be: 0: Event 1: Fault 2: Recovery	accessible-for-notify	current
hwStorageTrapEventID OID:1.3.6.1.4.1.2011.2.251.20.2.2.2	Counter64	Original alarm ID	accessible-for-notify	current
hwStorageTrapEvent-tLevel OID:1.3.6.1.4.1.2011.2.251.20.2.2.3	Gauge	Alarm severity. The value can be: **2**: Info **3**: Warning **5**: Major **6**: Critical	accessible-for-notify	current
hwStorageTrapEvent-Sequence OID:1.3.6.1.4.1.2011.2.251.20.2.2.4	Gauge	Alarm SN, namely, unique identifier of the alarm.	accessible-for-notify	current
hwStorageTrapEvent-Time OID:1.3.6.1.4.1.2011.2.251.20.2.2.5	Gauge	UTC time when the alarm was generated, unit: seconds.	accessible-for-notify	current
hwStorageTrapEvent-tRecoveryTime OID:1.3.6.1.4.1.2011.2.251.20.2.2.6	Gauge	UTC time when the alarm was cleared, unit: seconds.	accessible-for-notify	current
hwStorageTrapEvent-Parameter OID:1.3.6.1.4.1.2011.2.251.20.2.2.7	OCTETSTRING	Alarm location parameter. This parameter can be used to resolve alarm description. It is used with the alarm configuration file of the upper-layer network management system.	accessible-for-notify	current

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hwStorageTrapEventID32Bit OID:1.3.6.1.4.1.2011.2.251.20.2.2.8	Gauge	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwStorageTrapEventTimeStr OID:1.3.6.1.4.1.2011.2.251.20.2.2.9	OCTET STRING	Time when an alarm was generated. string format.	accessible-for-notify	current
hwStorageTrapEventRecoveryTimeStr OID:1.3.6.1.4.1.2011.2.251.20.2.2.10	OCTET STRING	Time when an alarm was cleared. string format.	accessible-for-notify	current

4.4 Storage Hardware Modules

hwInfoPortIBTable

This table is used to query information about InfiniBand ports.

This interface is discarded in OceanStor Dorado V6 6.1.0.

The index to this table is **hwInfoPortIBID**.

The OID of this table is 1.3.6.1.4.1.2011.2.251.22.16500. Value "4294967295" or all-F indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwInfoPortIBID OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.1	OCTET STRING	ID of an InfiniBand port, table index	read-only	current
hwInfoPortIBParentType OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.2	gauge	Parent type of an InfiniBand port. Possible values are: **207**: controller **208**: expansion board **209**: interface module	read-only	current

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hwInfoPortIBParentID OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.3	OCTET STRING	Parent ID of an InfiniBand port	read-only	current
hwInfoPortIBLocation OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.4	OCTET STRING	Location of an InfiniBand port	read-only	current
hwInfoPortIBHealthStatus OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.5	gauge	Health status of an InfiniBand port. For details, see **HEALTH_STATUS_E**.	read-only	current
hwInfoPortIBRunningStatus OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.6	gauge	Operating status of an InfiniBand port. For details, see **RUNNING_STATUS_E**.	read-only	current
hwInfoPortIBType OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.7	gauge	Logical type of an InfiniBand port. For details, see **PORT_LOGIC_TYPE_E**.	read-only	current
hwInfoPortIBWorkingRate OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.8	gauge	Working rate of an InfiniBand port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	read-only	current
hwInfoPortIBWWN OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.9	OCTET STRING	WWN of an InfiniBand port	read-only	current

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hwInfoPortIBRole OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.10	gauge	Role of an InfiniBand port. Possible values are: **2**: initiator **3**: target **4**: initiator and target **4294967295**: invalid	read-only	current
hwInfoPortIBSymbolError OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.11	gauge	Symbol error statistics of an InfiniBand port	read-only	current
hwInfoPortIBLinkErrorRecovery OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.12	gauge	Statistics of rectifying an InfiniBand port's connection errors	read-only	current
hwInfoPortIBReceiveErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.13	gauge	InfiniBand port receiving errors	read-only	current
hwInfoPortIBRemoteReceiveErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.14	gauge	InfiniBand port remotely receiving errors	read-only	current
hwInfoPortIBReceiveTransmitErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.15	gauge	InfiniBand port receiving and forwarding errors	read-only	current
hwInfoPortIBNotSendPacakges OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.16	gauge	Count of packets that are not sent by an InfiniBand port	read-only	current
hwInfoPortIBReceiveConstraintErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.17	gauge	InfiniBand port receiving constraint errors	read-only	current

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hwInfoPortIBLinkErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.18	gauge	Connection error of an InfiniBand port	read-only	current
hwInfoPortIBBufferOverrunErrors OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.19	gauge	Buffer overflow error of an InfiniBand port	read-only	current
hwInfoPortIBStartTime OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.20	OCTET STRING	Start time of an InfiniBand port	read-only	current
hwInfoPortIBMaxSpeed OID: 1.3.6.1.4.1.2011.2.251.22.16500.1.21	gauge	Maximum rate of an InfiniBand port (unit: Mbit/s)	read-only	current

4.5 TOPO Module

hwIsmAccessNodeTable

This table gives basis device information.

The index to this table is **hwIsmNENodeCode**.

The OID of this table is 1.3.6.1.4.1.2011.2.91.9.1.1.

Node	Data Type	Meaning	Permission	Status
hwIsmNENodeCode OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.1	OCTET STRING	ID of an NE node.	Read-only	Current
hwIsmNENodeType OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.2	Integer	Type of an NE node.	Read-only	Current

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hwIsmNENodeWorkingMode OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.3	Integer	Working mode of an NE node, **not used currently.**	Read-only	Current
hwIsmNENodeIPAddress OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.4	IpAddress	IP address of an NE node, **not used currently.**	Read-only	Current
hwIsmNENodeContextName OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.5	OCTET STRING	Name of the context.	Read-only	Current
hwIsmNENodeContextEngineID OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.6	OCTET STRING	Engine ID of the context, **not used currently.**	Read-only	Current
hwIsmNENodeClusterName OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.7	OCTET STRING	Name of the NE cluster, **not used currently.**	Read-only	Current
hwIsmNENodeRunningStatus OID: 1.3.6.1.4.1.2011.2.91.9.1.1.1.8	Integer	Running status of an NE: 1. The SNMP agent communicates correctly with the device. 2. The SNMP agent fails to communicate with the device. **Not used currently.**	Read-only	Current

4.6 SAN Module

hwInfoMappingViewTable

This table provides information about MappingViews.

The index to this table is **hwInfoMappingViewID**.

The OID of this table is 1.3.6.1.4.1.2011.2.251.23.245.

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Node	Data Type	Meaning	Permission	Status
hwInfoMappingViewID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.1	OCTET STRING	ID of an Mapping View.	Read-only	Current
hwInfoMappingView-Name OID: 1.3.6.1.4.1.2011.2.251.23.245.1.2	OCTET STRING	Name of Mapping View.	Read-only	Current
hwInfoMappingViewIn-bandCommand OID: 1.3.6.1.4.1.2011.2.251.23.245.1.3	Gauge	Whether to enable the command device 0:enable 1:disable	Read-only	Current
hwInfoMappingView-LunGroupID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.4	OCTET STRING	ID of Lun Group in Mapping View.	Read-only	Current
hwInfoMappingView-LunGroupName OID: 1.3.6.1.4.1.2011.2.251.23.245.1.5	OCTET STRING	Name of Lun Group in Mapping View.	Read-only	Current
hwInfoMappingVie-wHostGroupID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.6	OCTET STRING	ID of Host Group in Mapping View.	Read-only	Current
hwInfoMappingVie-wHostGroupName OID: 1.3.6.1.4.1.2011.2.251.23.245.1.7	OCTET STRING	Name of Lun Group in Mapping View.	Read-only	Current
hwInfoMappingView-PortGroupID OID: 1.3.6.1.4.1.2011.2.251.23.245.1.8	OCTET STRING	ID of Port Group in Mapping View.	Read-only	Current

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hwInfoMappingView-PortGroupName OID: 1.3.6.1.4.1.2011.2.251.23.245.1.9	OCTET STRING	Name of Port Group in Mapping View.	Read-only	Current

5 MIB References for Storage Nodes (Standard Enterprise Node: 34774)

5.1 Common Module
5.2 Alarm Module
5.3 Performance Statistics Module
5.4 Physical Storage Modules
5.5 Logical Storage Modules
5.6 Storage Space Modules
5.7 Storage Hardware Modules
5.8 NAS Information Query Module

5.1 Common Module

deviceId

Node	Data Type	Meaning	Permission	Status
deviceId OID: 1.3.6.1.4.1.34774.4.1.1.1	OCTET STRING	Device serial number, namely, unique identifier of the device.	Read-only	Current

deviceType

Node	Data Type	Meaning	Permission	Status

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deviceType OID: 1.3.6.1.4.1.34774.4.1.1.2	Integer	Type of the storage device. For details, see **PRODUCT_MODE_E**.	Read-only	Current

status

Node	Data Type	Meaning	Permission	Status
status OID: 1.3.6.1.4.1.34774.4.1.1.3	Integer	System operating status. For details, see **RUNNING_STATUS_E**.	Read-only	Current

usedCapacity

Node	Data Type	Meaning	Permission	Status
usedCapacity OID: 1.3.6.1.4.1.34774.4.1.1.4	Counter64	Used capacity of the storage device. The unit is MB.	Read-only	Current

totalCapacity

Node	Data Type	Meaning	Permission	Status
totalCapacity OID: 1.3.6.1.4.1.34774.4.1.1.5	Counter64	Total capacity of the storage device. The unit is MB.	Read-only	Current

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version

Node	Data Type	Meaning	Permission	Status
version OID: 1.3.6.1.4.1.34774.4.1.1.6	OCTET STRING	Device version.	Read-only	Current

5.2 Alarm Module

eventTable

This table gives device alarm information, which can be queried. This table is used exclusively for ISM management software and is not available to others.

The index to this table is **hwIsmEventSequence**.

The OID of this node is 1.3.6.1.4.1.34774.4.1.3.1.

Node	Data Type	Meaning	Permission	Status
hwIsmEventType OID: 1.3.6.1.4.1.34774.4.1.3.1.1.1	gauge	Alarm type. The value can be: **0**: eventAlarm **1**: faultAlarm **2**: resumeAlarm	Read-write	Current
hwIsmEventID OID: 1.3.6.1.4.1.34774.4.1.3.1.1.2	Counter64	ID of an alarm. It indicates a collection of alarms with the same category.	Read-only	Current
hwIsmEventLevel OID: 1.3.6.1.4.1.34774.4.1.3.1.1.3	gauge	Alarm severity. The value can be: **2**: Info **3**: Warning **5**: Major **6**: Critical	Read-write	Current
hwIsmEventSequence OID: 1.3.6.1.4.1.34774.4.1.3.1.1.4	gauge	Alarm SN, namely, unique identifier of the alarm.	Read-write	Current

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Node	Data Type	Meaning	Permission	Status
hwIsmEventTime OID: 1.3.6.1.4.1.34774.4.1.3.1.1.5	gauge	UTC time when the alarm was generated. The unit is second.	Read-write	Current
hwIsmEventRecoveryTime OID: 1.3.6.1.4.1.34774.4.1.3.1.1.6	gauge	UTC time when the alarm was cleared. The unit is second.	Read-write	Current
hwIsmEventParameter OID: 1.3.6.1.4.1.34774.4.1.3.1.1.7	OCTET STRING	Alarm location parameter. This parameter can be used to resolve alarm description. It is used with the alarm configuration file of the upper-layer network management system.	Read-write	Current
hwIsmEventRowStatus OID: 1.3.6.1.4.1.34774.4.1.3.1.1.20	Integer	Row status value. The value can be: 1: Row Active 6: Remove a alarm according to the alarm serial number.	Read-write	Current

eventType

This table is used exclusively for ISM management software and is not available to others.

The OID of this node is 1.3.6.1.4.1.34774.4.1.4.2.

Bound Variable	Data Type	Meaning	Permission	Status
hwIsmTrapEventType OID:1.3.6.1.4.1.34774.4.1.3.3.1	Gauge	Alarm type. The value can be: 0: Event 1: Fault 2: Recovery	accessible-for-notify	current
hwIsmTrapEventID OID:1.3.6.1.4.1.34774.4.1.3.3.2	Counter64	Original alarm ID	accessible-for-notify	current

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hwIsmTrapEventLevel OID:1.3.6.1.4.1.34774.4.1.3.3.3	Gauge	Alarm severity. The value can be: **2**: Info **3**: Warning **5**: Major **6**: Critical	accessible-for-notify	current
hwIsmTrapEventSequence OID:1.3.6.1.4.1.34774.4.1.3.3.4	Gauge	Alarm SN, namely, unique identifier of the alarm.	accessible-for-notify	current
hwIsmTrapEventTime OID:1.3.6.1.4.1.34774.4.1.3.3.5	Gauge	UTC time when the alarm was generated, unit: seconds.	accessible-for-notify	current
hwIsmTrapEventRecoveryTime OID:1.3.6.1.4.1.34774.4.1.3.3.6	Gauge	UTC time when the alarm was cleared, unit: seconds.	accessible-for-notify	current
hwIsmTrapEventParameter OID:1.3.6.1.4.1.34774.4.1.3.3.7	OCTET STRING	Alarm location parameter. This parameter can be used to resolve alarm description. It is used with the alarm configuration file of the upper-layer network management system.	accessible-for-notify	current
hwIsmTrapEventID32Bit OID:1.3.6.1.4.1.34774.4.1.3.3.8	Gauge	Alarm ID of I2000 network management software.	accessible-for-notify	current
hwIsmTrapEventTimeStr OID:1.3.6.1.4.1.34774.4.1.3.3.9	OCTET STRING	Time when an alarm was generated. string format.	accessible-for-notify	current
hwIsmTrapEventRecoveryTimeStr OID:1.3.6.1.4.1.34774.4.1.3.3.10	OCTET STRING	Time when an alarm was cleared. string format.	accessible-for-notify	current

forwardAddrTable

This table gives information about alarm forwarding address information, involving IP addresses, ports, and versions. This table is used exclusively for ISM management software and is not available to others.

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The index to this table is **forwardAddrIndex**.

The OID of this node is 1.3.6.1.4.1.34774.4.1.2.1.

Node	Data Type	Meaning	Permission	Status
forwardAddrIndex OID: 1.3.6.1.4.1.34774.4.1.2.1.1.1	OCTET STRING	Index to the alarm forwarding address. The value can be 0 to 3.	Read-write	Current
forwardAddrIP OID: 1.3.6.1.4.1.34774.4.1.2.1.1.2	IpAddress	IP address to which the alarm is forwarded. Only support IPv4 format.	Read-write	Current
forwardAddrPort OID:1 .3.6.1.4.1.34774.4.1.2.1.1.3	Gauge	Port of the alarm forwarding address. The value can be 1 to 65535.	Read-create	Current
forwardAddrTrapVer OID: 1.3.6.1.4.1.34774.4.1.2.1.1.4	Integer	Version of the alarm forwarding address. The value can be: ● 1: SNMPv1 ● 2: SNMPv2c ● 3: SNMPv3 When configuring the alarm forwarding address, if this field is not filled, the default is SNMPv2c.	Read-create	Current
forwardAddrRowStatus OID:1 .3.6.1.4.1.34774.4.1.2.1.1.5	Integer	Row status value. The value can be: ● 1: Modify the alarm forwarding address ● 4: Add the alarm forwarding address ● 6: Delete the alarm forwarding address	Read-create	Current
forwardAddrIPNew OID: 1.3.6.1.4.1.34774.4.1.2.1.1.6	OCTET STRING	New alarm forwarding address. Support IPv4 or IPv6 format. If **forwardAddrIP** and **forwardAddrIPNew** fields are filled with values, take the value of the **forwardAddrIPNew** field by default.	Read-create	Current

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forwardAddrTrapType OID: 1.3.6.1.4.1.34774.4.1.2.1.1.7	Integer	Type of the alarm forwarding address. The value can be: ● 1: Parsed. Sends parsed alarms to the alarm forwarding address (Trap OID:1.3.6.1.4.1.2011.2.91.10.2.1.0.1) ● 2: Original. Sends original alarms to the alarm forwarding address (Trap OID:1.3.6.1.4.1.34774.4.1.4.2). ● 3: All. Sends Parsed, Original, Parsed_time_string and Original_time_string alarms to the destination address for trap forwarding. ● 5: Parsed_time_string. Sends parsed alarms to the destination address for trap forwarding (Trap OID:1.3.6.1.4.1.2011.2.251.20.1.2.1) ● 6: Original_time_string. Sends original alarms to the destination address for trap forwarding (Trap OID:1.3.6.1.4.1.2011.2.251.20.2.1.1). ● 7: Parsed_time_string_precise. Sends parsed alarms to the destination address for trap forwarding. The alarm ID of each alarm corresponds to an independent OID, and different OIDs are used for alarm generation and clearance. For details, see **hwStorageAlarmReportingPrecise** in the *MIB Interface* and the *Alarm OID List (Only for Parsed_time_string_precise).xlsx*. When configuring the alarm forwarding address, if this field is not filled, the default is Original.		

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forwardAddrUSMU-serName OID: 1.3.6.1.4.1.34774.4.1.2.1.1.8	OCTET STRING	USM user name of the destination address for trap forwarding. When the trap forwarding address type is SNMPv3, this field is mandatory and users must ensure a specified USM user exists in the system. Otherwise, this operation fails to be performed. If this field is mandatory, and the parameter is not delivered or no value is specified for the parameter, the system default value is **Kaimse**.	Read-create	Current

5.3 Performance Statistics Module

hwPerformanceSwitch

This node indicates the status of the performance statistics switch.

Node	Data Type	Meaning	Permission	Status
hwPerformanceSwitch OID: 1.3.6.1.4.1.34774.4.1.21.1	Integer	Performance statistics switch. The value can be: 0: Disabled 1: Enabled	Read-write	Current

hwPerfNodeTable

This table is used to query the performance data of nodes.

The index to this table is **hwPerfNodeIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.3. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

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Node	Data Type	Meaning	Permission	Status
hwPerfNodeIfIndex OID: 1.3.6.1.4.1.34774.4.1.21.3.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwPerfNodeCPUUsage OID: 1.3.6.1.4.1.34774.4.1.21.3.1.2	Gauge	CPU usage of the node (unit: %).	Read-only	Current
hwPerfNodeAvgCacheUsage OID: 1.3.6.1.4.1.34774.4.1.21.3.1.3	Gauge	Average cache usage of the node (unit: %).	Read-only	Current
hwPerfNodeTotalIOPS OID: 1.3.6.1.4.1.34774.4.1.21.3.1.5	Gauge	Total IOPS of the node (unit: IO/s).	Read-only	Current
hwPerfNodeReadIOPS OID: 1.3.6.1.4.1.34774.4.1.21.3.1.6	Gauge	Read IOPS of the node (unit: IO/s).	Read-only	Current
hwPerfNodeWriteIOPS OID: 1.3.6.1.4.1.34774.4.1.21.3.1.7	Gauge	Write IOPS of the node (unit: IO/s).	Read-only	Current
hwPerfNodeTotalTraffic OID: 1.3.6.1.4.1.34774.4.1.21.3.1.8	Gauge	Total bandwidth for the node (unit: MB/s).	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwPerfNodeReadTraffic OID: 1.3.6.1.4.1.34774.4.1.21.3.1.9	Gauge	Read bandwidth for the node (unit: MB/s).	Read-only	Current
hwPerfNodeWriteTraffic OID: 1.3.6.1.4.1.34774.4.1.21.3.1.10	Gauge	Write bandwidth for the node (unit: MB/s).	Read-only	Current

hwPerfPortTable

This table is used to query the performance data of ETH, FC, FCoE, Bond ports.

The index to this table is **hwPerfPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.5. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwPerfPortIfIndex OID: 1.3.6.1.4.1.34774.4.1.21.5.1.1	Gauge	Index to the port.	Read-only	Current
hwPerfPortTotalIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.3	Gauge	IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortReadIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.4	Gauge	Read IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortWriteIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.5	Gauge	Write IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortTotalTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.6	Gauge	Total bandwidth for the port (unit: MB/s).	Read-only	Current

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hwPerfPortReadTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.7	Gauge	Read bandwidth for the port (unit: MB/s).	Read-only	Current
hwPerfPortWriteTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.8	Gauge	Write bandwidth for the port (unit: MB/s).	Read-only	Current
hwPerfPortMaxTraffic OID: 1.3.6.1.4.1.34774.4.1.21.5.1.9	Gauge	Maximum bandwidth for the port (unit: MB/s).	Read-only	Current
hwPerfPortMaxIOPS OID: 1.3.6.1.4.1.34774.4.1.21.5.1.10	Gauge	Maximum IOPS of the port (unit: IO/s).	Read-only	Current
hwPerfPortLocation OID: 1.3.6.1.4.1.34774.4.1.21.5.1.11	OCTET STRING	Location of the port.	Read-only	Current

hwPerfLunTable

This table is used to query the performance data (skip the grayed texts) of LUNs.

The index to this table is **hwPerfLunID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.4. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwPerfLunID OID:1.3.6.1.4.1.34774.4.1.21.4.1.1	Gauge	LUN ID.	Read-only	Current
hwPerfLunHitRate OID:1.3.6.1.4.1.34774.4.1.21.4.1.2	Gauge	I/O hit ratio of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunTotalIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.3	Gauge	Current IOPS of the LUN (unit: IO/s).	Read-only	Current

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hwPerfLunReadIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.4	Gauge	Current read IOPS of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunWriteIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.5	Gauge	Current write IOPS of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunTotalTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.6	Gauge	Current total bandwidth for the LUN (unit: MB/s).	Read-only	Current
hwPerfLunReadTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.7	Gauge	Current read bandwidth for the LUN (unit: MB/s).	Read-only	Current
hwPerfLunWriteTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.8	Gauge	Current write bandwidth for the LUN (unit: MB/s).	Read-only	Current
hwPerfLunReadIORate OID:1.3.6.1.4.1.34774.4.1.21.4.1.9	Gauge	Read I/O ratio of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunWriteIORate OID:1.3.6.1.4.1.34774.4.1.21.4.1.10	Gauge	Write I/O ratio of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunMaxTraffic OID:1.3.6.1.4.1.34774.4.1.21.4.1.11	Gauge	Maximum bandwidth for the LUN (unit: MB/s).	Read-only	Current
hwPerfLunMaxIOPS OID:1.3.6.1.4.1.34774.4.1.21.4.1.12	Gauge	Maximum IOPS of the LUN (unit: IO/s).	Read-only	Current
hwPerfLunAverageIOResponseTime OID:1.3.6.1.4.1.34774.4.1.21.4.1.13	Gauge	Average IO response time (unit:us).	Read-only	Current
hwPerfLunMaxIOResponseTime OID:1.3.6.1.4.1.34774.4.1.21.4.1.14	Gauge	Maximum IO response time (unit:us).	Read-only	Current
hwPerfLunAverageReadIOLatency OID:1.3.6.1.4.1.34774.4.1.21.4.1.15	Gauge	Average read IO response time (unit:us).	Read-only	Current

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hwPerfLunAverageWriteIOLatency OID:1.3.6.1.4.1.34774.4.1.21.4.1.16	Gauge	Average write IO response time (unit:us).	Read-only	Current

hwPerfCacheTable

This table is used to query the performance data of caches.

The index to this table is **hwPerfCacheID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.7. The value of performance statistics entry is "-1", indicates invalid.

Node	Data Type	Meaning	Permission	Status
hwPerfCacheID OID:1.3.6.1.4.1.34774.4.1.21.7.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwPerfCacheReadUtilization OID:1.3.6.1.4.1.34774.4.1.21.7.1.2	Integer32	Cache read usage (unit: %). **not used currently**.	Read-only	Current
hwPerfCacheWriteUtilization OID:1.3.6.1.4.1.34774.4.1.21.7.1.3	Integer32	Cache write usage (unit: %).	Read-only	Current
hwPerfCacheMirrorWriteUtilization OID:1.3.6.1.4.1.34774.4.1.21.7.1.4	Integer32	Cache mirror write usage (unit: %). **not used currently**.	Read-only	Current

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hwPerfCacheHitRatio OID:1.3.6.1.4.1.34774.4.1.21.7.1.5	Integer32	Cache hit ratio.	Read-only	Current

hwPerfControllerNFSV3Table

This table is used to query the NFSv3 performance data of controllers.

The index to this table is **hwPerfControllerNFSV3ID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.21.10. The value "4294967295" or "F" of performance statistics entry indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwPerfControllerNFSV3ID OID: 1.3.6.1.4.1.34774.4.1.21.10.1.1	OCTET STRING	Controller ID.	Not-accessible	Current
hwPerfControllerNFSV3OPS OID: 1.3.6.1.4.1.34774.4.1.21.10.1.2	Counter64	NFSv3 OPS of the controller	Read-only	Current
hwPerfControllerNFSV3Bps OID: 1.3.6.1.4.1.34774.4.1.21.10.1.3	Counter64	NFSv3 throughput of the controller (unit: byte/s)	Read-only	Current

5.4 Physical Storage Modules

Note: If the value of an interface parameter is 4294967295 or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.

hwStorageFCPortTable

This table provides information about FC ports.

The index to this table is **hwStorageFCPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.7.

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Node	Data Type	Meaning	Permission	Status
hwStorageFCPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.1	Gauge	Index to the FC port.	Read-only	Current
hwStorageFCPortBoardIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.2	Gauge	Index to the board of the FC port.	Read-only	Current
hwStorageFCPortID OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.3	Gauge	ID of the FC port.	Read-only	Current
hwStorageFCPortStatus OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.4	Integer	Status of the FC port. The value can be: **0**: Disconnected **1**: Connected **2**: Faulty	Read-only	Current
hwStorageFCPortConfigRate OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.5	Integer	Working Rate of the FC port, (unit: 1000Mbit/s). The value can be: **0**: Auto **1**: GBPS-1 **2**: GBPS-2 **4**: GBPS-4 **8**: GBPS-8 **16**: GBPS-16 The Current rate can be **1 Gbit/s**, **2 Gbit/s**, **4 Gbit/s**, **8 Gbit/s**, or **16 Gbit/s**. The conversion unit is 1000. **0** indicates auto-negotiation. **-1** indicates invalid.	Read-only	Current
hwStorageFCPortMode OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.6	Integer	Mode of the FC port. The value can be: ● -1: Invalid ● 0: fabric ● 1: FC-AL ● 2: P2P ● 3: Auto	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageFCPortWWN OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.7	OCTET STRING	WWN of the FC port.	Read-only	Current
hwStorageFCPortLogicType OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.8	Integer	Logical type of FC port. The value can be: ● 1: service ● 2: ibc ● 3: ibs ● 4: expansion ● 5: management ● 6: upsmanagement ● 8: maintenance	Read-only	Current
hwStorageFCPortSpeedRate OID:1.3.6.1.4.1.34774.4.1.19.8.7.1.9	Integer	Speed rate of FC port(unit: Mbit/s). If the parameter value is **-1**, this parameter is invalid.	Read-only	Current

hwStorageSASPortTable

This table provides information about SAS ports.

The index to this table is **hwStorageSASPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.8.

Node	Data Type	Meaning	Permission	Status
hwStorageSASPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.1	Gauge	Index to the SAS port.	Read-only	Current
hwStorageSASPortBoardIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.2	Gauge	Index to the board of the SAS port.	Read-only	Current
hwStorageSASPortID OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.3	Gauge	ID of the SAS port.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageSASPort-Status OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.4	Integer	Status of the SAS port. The value can be: **0**: disconnected **1**: connected **2**: faulty	Read-only	Current
hwStorageSASPort-ConfigRate OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.5	Integer	Rate of the SAS port. (unit: 1000 Mbit/s). If the parameter value is **-1**, this parameter is invalid.	Read-only	Current
hwStorageSASPortWWN OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.6	OCTET STRING	WWN of the SAS port.	Read-only	Current
hwStorageSASPortLogicType OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.7	Integer	Logical type of SAS port. The value can be: ● 1: service ● 2: ibc ● 3: ibs ● 4: expansion ● 5: management ● 6: upsmanagement ● 8: maintenance	Read-only	Current
hwStorageSASPort-SpeedRate OID:1.3.6.1.4.1.34774.4.1.19.8.8.1.8	Integer	Speed rate of SAS port. The value can be: ● 1: Gbps15 ● 2: Gbps30 ● 3: Gbps60 ● 4: Gbps12	Read-only	Current

hwStorageISCSIPortTable

This table provides information about iSCSI ports.

The index to this table is **hwStorageISCSIPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.9.

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Node	Data Type	Meaning	Permission	Status
hwStorageISCSIPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.1	Gauge	Index to the iSCSI port.	Read-only	Current
hwStorageISCSIPortBoardIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.2	Gauge	Index to the board of the iSCSI port.	Read-only	Current
hwStorageISCSIPortID OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.3	Gauge	ID of the iSCSI port.	Read-only	Current
hwStorageISCSIPortStatus OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.4	Integer	Status of the iSCSI port. The value can be: 0: disconnected 1: connected 2: faulty	Read-only	Current
hwStorageISCSIPortIP OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.5	IpAddress	IP address of the iSCSI port.	Read-only	Current
hwStorageISCSIPortNetMask OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.6	IpAddress	Mask of the iSCSI port.	Read-only	Current
hwStorageISCSIPortBindMode OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.7	Integer	Bind mode of the iSCSI port, **not used currently**.	Read-only	Current
hwStorageISCSIPortLogicType OID:1.3.6.1.4.1.34774.4.1.19.8.9.1.8	Integer	Logical type of ISCSI port. The value can be: ● 1: service ● 2: ibc ● 3: ibs ● 4: expansion ● 5: management ● 6: upsmanagement ● 8: maintenance	Read-only	Current

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hwStorageFrontEndHostPortTable

This table provides basic information about the SAS, FC, iSCSI ports whose performance statistics can be collected.

The index to this table is **hwStorageFrontEndHostPortIfIndex**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.11.

Node	Data Type	Meaning	Permission	Status
hwStorageFrontEndHostPortIfIndex OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.1	Gauge	Combination index to the host ports.	Read-only	Current
hwStorageFrontEndHostPortType OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.2	Integer	Physical type of a host port: • 0: SAS • 1: FC • 5: iSCSI	Read-only	Current
hwStorageFrontEndHostPortStatus OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.3	Integer	Connection state of a host port: • 0: link down • 1: link up • 2: not exist • 3: link bad • 4: hardware error • 5: speed no match • 6: enable • 7: disable • 8: mac error • 9: lane no match	Read-only	Current
hwStorageFrontEndHostPortPhysAddress OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.4	OCTET STRING	IP address of a host port. If "--" is displayed, the value is invalid.	Read-only	Current
hwStorageFrontEndHostPortDescription OID:1.3.6.1.4.1.34774.4.1.19.8.11.1.5	OCTET STRING	Description of a host port.	Read-only	Current

hwStorageControllerTable

This table provides basic information about the controller.

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The index to this table is **hwStorageControllerID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.8.12.

Node	Data Type	Meaning	Permission	Status
hwStorageControllerID OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwStorageControllerName OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.2	OCTET STRING	Controller name	Read-only	Current
hwStorageControllerLocation OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.3	OCTET STRING	Controller location	Read-only	Current
hwStorageControllerHealthStatus OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.4	Integer	Controller health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwStorageControllerRunningStatus OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.5	Integer	Controller running status. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwStorageControllerSoftVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.6	OCTET STRING	Controller software version	Read-only	Current

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hwStorageController-Temperature OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.7	Integer	Controller temperature, unit: degree, not used currently.	Read-only	Current
hwStorageControllerIs-Master OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.8	Gauge	The state of controller, master or slave 0: Slave 1: Master Not used currently.	Read-only	Current
hwStorageControllerELabel OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.9	OCTET STRING	Controller electronic label	Read-only	Current
hwStorageControllerPCBVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.10	OCTET STRING	Controller PCB version	Read-only	Current
hwStorageControllerBMCVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.11	OCTET STRING	Controller BMC version. When the value is empty or "--", indicates unknown or invalid.	Read-only	Current
hwStorageControllerLogicVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.12	OCTET STRING	Controller logical version	Read-only	Current
hwStorageController-BIOSVer OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.13	OCTET STRING	Controller BIOS version	Read-only	Current
hwStorageControllerMemorySize OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.14	Gauge	Controller memory size, unit: MB	Read-only	Current
hwStorageControllerCPUInfo OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.15	OCTET STRING	CPU information, composed by four fields: manufacturer, make, model, and frequency.	Read-only	Current
hwStorageControllerVoltage OID:1.3.6.1.4.1.34774.4.1.19.8.12.1.16	Gauge	Controller voltage, unit: 0.1 V.	Read-only	Current

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5.5 Logical Storage Modules

*Note: If the value of an interface parameter is **4294967295** or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.*

hwStorageLunTable

This table is used to query the LUN information about a device.

The index to this table is **hwStorageLunID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.9.4.

Node	Data Type	Meaning	Permission	Status
hwStorageLunID OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.1	Gauge	ID of the LUN.	Read-only	Current
hwStorageLunName OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.2	OCTET STRING	Name of the LUN.	Read-only	Current
hwStorageLunWWN OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.3	OCTET STRING	WWN of the LUN.	Read-only	Current
hwStorageLunPoolID OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.4	Gauge	ID of the storage pool to which the LUN belongs.	Read-only	Current
hwStorageLunCapacity OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.5	Counter 64	Capacity of the LUN. The unit is KB.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageLunOwning-Controller OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.6	Gauge	Node id of the owning controller of the LUN. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwStorageLunStripDepth OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.7	Gauge	Stripe depth of the LUN. **Not used currently.**	Read-only	Current
hwStorageLunWriteS-trategy OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.8	Integer	Write policy of the LUN. The value can be: ● 1: Write through ● 2: Write back no mirroring ● 3: Write back mirroring ● 4: Write back mandatory nomirroring ● 5: Write back mandatory mirroring	Read-only	Current
hwStorageLunPrefetch-Strategy OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.9	Integer	Prefetch policy of the LUN. The value can be: ● 0: Non-prefetch ● 1: Constant prefetch ● 2: Variable prefetch ● 3: Intelligent prefetch	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwStorageLunPrefetch-Value OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.10	Gauge	Cache prefetch value of the LUN. The cache prefech policy can be: **1**. constant prefetch (unit: KB) **2**. variable prefetch (unit: times) **3**. non-prefetch or intelligent prefetch. If **4294967295** is returned, the value is invalid.	Read-only	Current
hwStorageLunStatus OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.11	Integer	Status of the LUN. The value can be: • 1: Normal • 2: Faulty • 3: Unformatted • 4: Formatting	Read-only	Current
hwStorageLunRowStatus OID:1.3.6.1.4.1.34774.4.1.19.9.4.1.12	Integer	Row status. • 1: Active	Read-write	Current

hwStorageCacheTable

This table provides cache information such as cache ID, cache size, cache high water level, and cache low water level.

The index to this table is **hwStorageCacheID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.19.9.1.

Node	Data Type	Meaning	Permission	Status

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hwStorageCacheID OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.1	Gauge	Controller node ID. Controller node ID = Controller enclosure ID x Number of controllers in the controller enclosure + Controller ID. In controller name **1A**, **1** indicates the controller enclosure ID and **A** indicates the controller ID (A, B, C, D correspond to 0, 1, 2, 3, respectively). If a controller enclosure has two controllers, the node ID is **2**. If a controller enclosure has four controllers, the node ID is **4**.	Read-only	Current
hwStorageCacheTotal-Capacity OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.2	Gauge	Cache size (unit: MB).	Read-only	Current
hwStorageCacheHigh-WaterLevel OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.3	Gauge	Cache high water level.	Read-only	Current
hwStorageCacheLow-WaterLevel OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.4	Gauge	Cache low water level.	Read-only	Current
hwStorageCacheRow-Status OID:1.3.6.1.4.1.34774.4.1.19.9.1.1.5	Integer	Row status. • 1: Active **not used currently.**	Read-write	Current

5.6 Storage Space Modules

Note: If the value of an interface parameter is 4294967295 or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.

hwInfoDiskDomainTable

This table is used to check the disk domain information about a device. The index to this table is **hwInfoDiskDomainID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.1.

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskDomainID OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.1	OCTET STRING	Disk domain ID, table index	Read-only	Current
hwInfoDiskDomainName OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.2	OCTET STRING	Name of a disk domain	Read-only	Current
hwInfoDiskDomainHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.3	Gauge	Health status of a disk domain. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoDiskDomainRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.4	Gauge	Running status of a disk domain. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoDiskDomainTotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.5	Counter64	Total capacity of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainFreeCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.6	Counter64	Free capacity of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainHotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.7	Counter64	Hot spare capacity of a disk domain. (unit: MB)	Read-only	Current
hwInfoDiskDomainUsedHotSpareCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.1.1.8	Counter64	Used hot spare capacity of a disk domain. (unit: MB)	Read-only	Current

hwInfoStoragePoolTable

This table is used to check the storage pool information about a device. The index to this table is **hwInfoStoragePoolID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.2.

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Node	Data Type	Description	Permission	Status
hwInfoStoragePoolID OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.1	OCTET STRING	ID of a storage pool, table index	Read-only	Current
hwInfoStoragePoolName OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.2	OCTET STRING	Name of a storage pool	Read-only	Current
hwInfoStoragePoolDiskDomainID OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.3	OCTET STRING	ID of the disk domain where a storage pool resides	Read-only	Current
hwInfoStoragePoolDiskDomainName OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.4	OCTET STRING	Name of the disk domain where a storage pool resides	Read-only	Current
hwInfoStoragePoolHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.5	Gauge	Health status of a storage pool. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoStoragePoolRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.6	Gauge	Operating status of a storage pool. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoStoragePoolTotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.7	Counter64	Total capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolSubscribedCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.8	Counter64	Total subscribed capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolFreeCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.9	Counter64	Available capacity of a storage pool (unit: MB)	Read-only	Current
hwInfoStoragePoolProtectionCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.10	Counter64	Data protection capacity of a storage pool (unit: MB)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoStoragePoolFull-Threshold OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.14	Gauge	Threshold of used capacity of a storage pool (unit: %)	Read-only	Current
hwInfoStoragePoolLunTotalCapacity OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.23	Counter64	Lun capacity of a storage pool (unit: MB)	read-only	current
hwInfoStoragePoolDataReductionRatio OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.24	OCTET STRING	Data Reduction Ratio **not used currently**.	read-only	current
hwInfoStoragePoolDeduplicationRatio OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.25	OCTET STRING	Data Deduplication Ratio **not used currently**.	read-only	current
hwInfoStoragePoolCompressionRatio OID:1.3.6.1.4.1.34774.4.1.23.4.2.1.26	OCTET STRING	Data Compression Ratio **not used currently**.	read-only	current

hwInfoPortGroupTable

This table is used to check the port group information about a device. The index to this table is **hwInfoPortGroupID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.4.

Node	Data Type	Description	Permission	Status
hwInfoPortGroupID OID:1.3.6.1.4.1.34774.4.1.23.4.4.1.1	OCTETSTRING	ID of a port group, table index	Read-only	Current
hwInfoPortGroupName OID:1.3.6.1.4.1.34774.4.1.23.4.4.1.2	OCTETSTRING	Name of a port group	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortGroupPortList OID:1.3.6.1.4.1.34774.4.1.23.4.4.1.3	OCTET STRING	List of ports in a port group. The format of list is: ID1: Type1: Location1, ID2: Type2: Location2…. The value of Type is: • 213: Port ETH • 212: Port FC • 252: Port FCoE	Read-only	Current

hwInfoHostTable

This table is used to check the host information about a device. The index to this table is **hwInfoHostID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.5.

Node	Data Type	Description	Permission	Status
hwInfoHostID OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.1	OCTET STRING	Host ID, table index	Read-only	Current
hwInfoHostName OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.2	OCTET STRING	Host name	Read-only	Current
hwInfoHostLocation OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.3	OCTET STRING	Host location	Read-only	Current
hwInfoHostHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.4	Gauge	Host health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoHostRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.5	Gauge	Host operating status. For details, see **RUNNING_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoHostOperating-System OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.6	Gauge	Host operating system. Possible values are: • 0: Linux • 1: Windows • 2: Solaris • 3: HP-UX • 4: AIX • 5: XenServer • 6: Mac OS • 7: VMware ESX • 8: VIS6000 • 9: Windows Server 2012 • 10: Oracle_VM • 11: OpenVMS • 12: Oracle_VM_Server_for_x86 • 13: Oracle_VM_Server_for_SPARC	Read-only	Current
hwInfoHostIPAddress OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.7	OCTET STRING	Host IP address	Read-only	Current
hwInfoHostNetwor-kName OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.8	OCTET STRING	Host domain name	Read-only	Current
hwInfoHostModel OID:1.3.6.1.4.1.34774.4.1.23.4.5.1.9	OCTET STRING	Host model	Read-only	Current

hwInfoHostGroupTable

This table is used to check the host group information about a device. The index to this table is **hwInfoHostGroupID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.6.

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Node	Data Type	Description	Permission	Status
hwInfoHostGroupID OID:1.3.6.1.4.1.34774.4.1.23.4.6.1.1	OCTET STRING	ID of a host group, table index	Read-only	Current
hwInfoHostGroupName OID:1.3.6.1.4.1.34774.4.1.23.4.6.1.2	OCTET STRING	Name of a host group	Read-only	Current
hwInfoHostGroupHost-List OID:1.3.6.1.4.1.34774.4.1.23.4.6.1.3	OCTET STRING	List of hosts in a host group. The format of list is: ID1: Name1, ID2: Name2…	Read-only	Current

hwInfoLunGroupTable

This table is used to check the LUN group information about a device. The index to this table is **hwInfoLunGroupID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.7.

Node	Data Type	Description	Permission	Status
hwInfoLunGroupID OID:1.3.6.1.4.1.34774.4.1.23.4.7.1.1	OCTET STRING	ID of a LUN group, table index	Read-only	Current
hwInfoLunGroupName OID:1.3.6.1.4.1.34774.4.1.23.4.7.1.2	OCTET STRING	Name of a LUN group	Read-only	Current
hwInfoLunGroupLunList OID:1.3.6.1.4.1.34774.4.1.23.4.7.1.3	OCTET STRING	List of LUNs in a LUN group. The format of list is: ID1: Name1, ID2: Name2…	Read-only	Current

hwInfoLunTable

This table is used to check the LUN information about a device. The index to this table is **hwInfoLunID**.

The prefix to the OID of this table is 1.3.6.1.4.1.34774.4.1.23.4.8.

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Node	Data Type	Description	Permission	Status
hwInfoLunID OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.1	OCTET STRING	LUN ID	Read-only	Current
hwInfoLunName OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.2	OCTET STRING	LUN name	Read-only	Current
hwInfoLunPoolID OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.3	OCTET STRING	ID of the storage pool to which the LUN belongs	Read-only	Current
hwInfoLunPoolName OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.4	OCTET STRING	Name of the storage pool to which the LUN belongs	Read-only	Current
hwInfoLunCapacity OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.5	Counter64	Total capacity of the LUN (unit: MB)	Read-only	Current
hwInfoLunSubscribedCapacity OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.6	Counter64	Capacity usage quota of the LUN (unit: MB)	Read-only	Current
hwInfoLunProtectionCapacity OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.7	Counter64	Data protection capacity quota of the LUN (unit: KB)	Read-only	Current
hwInfoLunSectorSize OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.8	Gauge	Sector size of the LUN (unit: B)	Read-only	Current
hwInfoLunHealthStatus OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.9	Gauge	Health status of the LUN. For details, see **HEALTH_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLunRunningStatus OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.10	Gauge	Operating status of the LUN. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoLunType OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.11	Gauge	LUN type. The value can be: **0**: thick LUN **1**: thin LUN	Read-only	Current
hwInfoLunIOPriority OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.12	Gauge	I/O priority of the LUN. The value can be: **1**: low **2**: middle **3**: high	Read-only	Current
hwInfoLunWWN OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.13	OCTET STRING	WWN of the LUN	Read-only	Current
hwInfoLunExposedToInitiator OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.14	Gauge	Whether the LUN is mapped to an initiator. The value can be: **0**: no **1**: yes	Read-only	Current
hwInfoLunWritePolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.15	Gauge	Cache write policy of the LUN. The value can be: **1**: write back **2**: write through **3**: forcible write back **4**: hybrid	Read-only	Current
hwInfoLunRunningWritePolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.16	Gauge	Operating cache write policy of the LUN. The value can be: **1**: write back **2**: write through **3**: forcible write back **4**: hybrid	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLunPrefetchPolicy OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.17	Gauge	Cache prefetch policy of the LUN. The value can be: **0**: non-prefetch **1**: constant prefetch **2**: variable prefetch **3**: intelligent prefetch	Read-only	Current
hwInfoLunPrefetchValue OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.20	Gauge	Cache prefetch value of the LUN. The cache prefech policy can be: **1**. constant prefetch (unit: KB) **2**. variable prefetch (unit: times) **3**. non-prefetch or intelligent prefetch. If **4294967295** is returned, the value is invalid.	Read-only	Current
hwInfoLunOwnerController OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.21	OCTET STRING	Owning controller of the LUN	Read-only	Current
hwInfoLunWorkController OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.22	OCTET STRING	Working controller of the LUN	Read-only	Current
hwInfoLunIsAddToLunGroup OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.25	Gauge	Whether the LUN is added to the LUN group. The value can be: **0**: no **1**: yes	Read-only	Current
hwInfoLunDIFSwitch OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.26	Gauge	Whether the DIF function is enabled for the LUN. The value can be: **0**: no **1**: yes	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLunRemote-LUNWWN OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.27	OCTET STRING	WWN of the remote LUN	Read-only	Current
hwInfoLunUsageType OID: 1.3.6.1.4.1.34774.4.1.23.4.8.1.28	Gauge	LUN type. The value can be: **0**: internal LUN **1**: external LUN **2**: VVOL LUN **3**: PE LUN **5**: global image repository LUN **6**: local image repository LUN **7**: container configuration LUN	Read-only	Current

5.7 Storage Hardware Modules

Note: If the value of an interface parameter is 4294967295 or "F" (parameter type is unsigned), "--" or empty (parameter type is a character string), it indicates that the parameter is unknown or invalid.

hwInfoDiskTable

This table is used to check the disk information about a device. The index to this table is **hwInfoDiskID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.1.

Node	Data Type	Meaning	Permission	Status
hwInfoDiskID OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.1	OCTET STRING	Disk ID, table index	Read-only	Current
hwInfoDiskHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.2	Gauge	Disk health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.3	Gauge	Disk running status. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoDiskLocation OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.4	OCTET STRING	Disk location	Read-only	Current
hwInfoDiskType OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.5	Gauge	Disk type. Possible values are: • 0: FC • 1: SAS • 2: SATA • 3: SSD • 4: NL_SAS • 5: SLC SSD • 6: MLC SSD • 7:FC_SED • 8:SAS_SED • 9:SATA_SED • 10:SSD_SED • 11:NL_SAS_SED • 12:SLC_SSD_SED • 13:MLC_SSD_SED • 14:NVMe_SSD • 15:SMR_SATA • 16:NVMe_SSD_SED • 17:SCM_SSD • 18:SCM_SED	Read-only	Current
hwInfoDiskCapacity OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.6	Counter64	Disk capacity(unit: MB)	Read-only	Current
hwInfoDiskRole OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.7	Gauge	Logical type of a disk. Possible values are: • 1: FREE • 2: MEMBER • 3: SPARE • 4: CACHE	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskSpeed OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.8	Gauge	Rotational speed of a disk (unit: rpm)	Read-only	Current
hwInfoDiskInterface-Bandwidth OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.9	Gauge	Disk interface bandwidth (unit: Mbit/s)	Read-only	Current
hwInfoDiskSectorSize OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.10	Gauge	Sector size of a disk (unit: B)	Read-only	Current
hwInfoDiskTemperature OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.11	Integer	Disk temperature (unit: °C)	Read-only	Current
hwInfoDiskModel OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.12	OCTET STRING	Disk model	Read-only	Current
hwInfoDiskFirmwareVersion OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.13	OCTET STRING	Firmware version of a disk	Read-only	Current
hwInfoDiskManufacturer OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.14	OCTET STRING	Disk manufacturer	Read-only	Current
hwInfoDiskSerialNumber OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.15	OCTET STRING	Disk SN	Read-only	Current
hwInfoDiskLightStatus OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.16	Gauge	Disk indicator status. Possible values are: • 0: OFF • 1: ON • 2: STEADY ON	Read-only	Current
hwInfoDiskDiskDomainID OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.17	OCTET STRING	ID of a disk domain	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoDiskDiskDomainName OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.18	OCTET STRING	Name of a disk domain	Read-only	Current
hwInfoDiskDiskDomainTierID OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.19	OCTET STRING	ID of a disk domain tier	Read-only	Current
hwInfoDiskCofferDisk OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.20	Gauge	To determine whether a disk is a coffer disk. Possible values are: • 0: NO • 1: YES	Read-only	Current
hwInfoDiskRunTime OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.21	Gauge	Disk operation period (unit: day)	Read-only	Current
hwInfoDiskProgress OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.22	Gauge	Progress of disk reconstruction, copyback, pre-copy, or destruction (unit: %)	Read-only	Current
hwInfoDiskBarCode OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.23	OCTET STRING	Disk barcode	Read-only	Current
hwInfoDiskCapacityUsage OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.24	Gauge	Capacity usage of a disk. (unit: %).	Read-only	Current
hwInfoDiskHealthMark OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.25	Gauge	Health score of a disk. If the value is 255, indicating invalid	Read-only	Current
hwInfoDiskElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.1.1.26	OCTET STRING	Electronic label of a disk	Read-only	Current

hwInfoControllerTable

This table is used to check the controller information about a device. The index to this table is **hwInfoControllerID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.2.

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Node	Data Type	Meaning	Permission	Status
hwInfoControllerID OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.1	OCTET STRING	Controller ID, table index	Read-only	Current
hwInfoControllerHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.2	Gauge	Controller health status. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoControllerRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.3	Gauge	Controller running status. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoControllerCPU OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.4	OCTET STRING	Controller CPU	Read-only	Current
hwInfoControllerLocation OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.5	OCTET STRING	Controller location	Read-only	Current
hwInfoControllerRole OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.6	Gauge	Controller role. Possible values are: ● 0: Normal ● 1: Master ● 2: Slave	Read-only	Current
hwInfoControllerCacheCapacity OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.7	Gauge	Cache capacity of a controller (unit: MB)	Read-only	Current
hwInfoControllerCPUUsage OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.8	Gauge	CPU usage of a controller (unit: %)	Read-only	Current
hwInfoControllerMemoryUsage OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.9	Gauge	Memory usage of the operating system in the controller (unit: %)	Read-only	Current
hwInfoControllerVoltage OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.10	Gauge	Power voltage of a controller (unit: 0.1 V)	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoControllerSoftwareVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.11	OCTET STRING	Software version of a controller	Read-only	Current
hwInfoControllerPCBVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.12	OCTET STRING	PCB version of a controller	Read-only	Current
hwInfoControllerSESVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.13	OCTET STRING	SES version of a controller	Read-only	Current
hwInfoControllerBMCVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.14	OCTET STRING	BMC version of a controller. When the value is empty or "--", indicates unknown or invalid.	Read-only	Current
hwInfoControllerLogicVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.15	OCTET STRING	Logical version of a controller	Read-only	Current
hwInfoControllerBIOSVersion OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.16	OCTET STRING	BIOS version of a controller	Read-only	Current
hwInfoControllerElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.2.1.17	OCTET STRING	Electronic label of a controller	Read-only	Current

hwInfoPowerTable

This table is used to check the power supply information about a device. The index to this table is **hwInfoPowerID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.3. When the value is empty or "--", indicates unknown or invalid.

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Node	Data Type	Meaning	Permission	Status
hwInfoPowerID OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.1	OCTET STRING	ID of a power supply, table index	Read-only	Current
hwInfoPowerLocation OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.2	OCTET STRING	Location of a power supply	Read-only	Current
hwInfoPowerHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.3	Gauge	Health status of a power supply. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPowerRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.4	Gauge	Operating status of a power supply. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPowerType OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.5	Gauge	Type of a power supply. Possible values are: • 0: DC • 1: AC • 2: HV • 255:Unknown	Read-only	Current
hwInfoPowerManufacturer OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.6	OCTET STRING	Manufacturer of a power supply	Read-only	Current
hwInfoPowerModle OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.7	OCTET STRING	Model of a power supply	Read-only	Current
hwInfoPowerVersion OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.8	OCTET STRING	Version of a power supply	Read-only	Current
hwInfoPowerProduceDate OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.9	OCTET STRING	Production date of a power supply	Read-only	Current
hwInfoPowerSerialNumber OID:1.3.6.1.4.1.34774.4.1.23.5.3.1.10	OCTET STRING	SN of a power supply	Read-only	Current

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hwInfoFanTable

This table is used to check the fan information about a device. The index to this table is **hwInfoFanID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.4. When the value is empty or "--", indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwInfoFanID OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.1	OCTET STRING	Fan ID, table index	Read-only	Current
hwInfoFanLocation OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.2	OCTET STRING	Fan Location	Read-only	Current
hwInfoFanHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.3	Gauge	Health status of a fan. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoFanRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.4	Gauge	Operating status of a fan. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoFanRunningLevel OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.5	Gauge	Operating level of a fan. Possible values are: • 0: Low • 1: Normal • 2: High	Read-only	Current
hwInfoFanElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.4.1.6	OCTET STRING	Electronic label of a fan	Read-only	Current

hwInfoBBUTable

This table is used to check the BBU information about a device. The index to this table is **hwInfoBBUID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.5.

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Node	Data Type	Meaning	Permission	Status
hwInfoBBUID OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.1	OCTET STRING	BBU ID, table index	Read-only	Current
hwInfoBBULocation OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.2	OCTET STRING	BBU Location	Read-only	Current
hwInfoBBUHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.3	Gauge	Health status of a BBU. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoBBURunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.4	Gauge	Running status of a BBU. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoBBUType OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.5	Gauge	Type of a BBU. Possible values are: ● 0: BBU	Read-only	Current
hwInfoBBUCurrentVoltage OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.6	Gauge	Current voltage of a BBU (unit: 0.1 V)	Read-only	Current
hwInfoBBUNumberOfDischarges OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.7	Gauge	Discharge times of a BBU (unit: times)	Read-only	Current
hwInfoBBUFirmwareVersion OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.8	OCTET STRING	Firmware version of a BBU	Read-only	Current
hwInfoBBUDeliveredOn OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.9	OCTET STRING	Shipping date of a BBU	Read-only	Current
hwInfoBBUOwningController OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.10	OCTET STRING	Owning controller of a BBU	Read-only	Current
hwInfoBBUElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.5.1.11	OCTET STRING	Electronic label of a BBU	Read-only	Current

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hwInfoEnclosureTable

This table is used to check the enclosure information about a device. The index to this table is **hwInfoEnclosureID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.6. When the value is empty or "--", indicates unknown or invalid.

Node	Data Type	Meaning	Permission	Status
hwInfoEnclosureID OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.1	OCTET STRING	Enclosure ID, table index	Read-only	Current
hwInfoEnclosureName OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.2	OCTET STRING	Enclosure name	Read-only	Current
hwInfoEnclosureLogicType OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.3	Gauge	Logical type of an enclosure. Possible values are: ● 0: EXP ● 1: CTRL ● 2: DSW ● 3: MSW ● 4: SVP	Read-only	Current
hwInfoEnclosureHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.4	Gauge	Health status of an enclosure. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoEnclosureRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.5	Gauge	Operating status of an enclosure. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoEnclosureLocation OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.6	OCTET STRING	Enclosure location	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
hwInfoEnclosureType OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.7	Gauge	Enclosure type. Possible values are: 0:BMC Controller 1:2 U 2-controller 12-slot 3.5-inch 6 Gbit/s SAS controller enclosure 2:2 U 2-controller 24-slot 2.5-inch 6 Gbit/s SAS controller enclosure 16:2 U 12-slot 3.5-inch 6 Gbit/s SAS disk enclosure 17:2 U 24-slot 2.5-inch 6 Gbit/s SAS disk enclosure 18:4 U 24-slot 3.5-inch 6 Gbit/s SAS disk enclosure 19:4 U 24 Slot 3.5 FC Disks Enclosure 20:1 U PCIE Switch Enclosure 21:4 U 75-slot 3.5-inch 6 Gbit/s SAS disk enclosure 22:Service Processor Enclosure 23:2 U 2-controller 12-slot 3.5-inch 6 Gbit/s SAS controller enclosure 24:2 U 25-slot 2.5-inch 6 Gbit/s SAS disk enclosure 25:4 U 24-slot 3.5-inch 6 Gbit/s SAS disk enclosure 26:2 U 2-controller 25-slot 2.5-inch 6 Gbit/s SAS controller enclosure 37:2 U 2-controller 12-slot 3.5-inch 6 Gbit/s SAS controller enclosure 38:2 U 2-controller 25-slot 2.5-inch 6 Gbit/s SAS controller enclosure 39:4 U 75-slot 3.5-inch 12 Gbit/s SAS disk enclosure 40:2 U 2-controller 25-slot 2.5-inch 12 Gbit/s SAS controller enclosure	Read-only	Current

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Node	Data Type	Meaning	Permission	Status
		65:2 U 25-slot 2.5-inch 12 Gbit/s SAS disk enclosure 66:4 U 24-slot 3.5-inch 12 Gbit/s SAS disk enclosure 67:2 U 25-slot 2.5-inch SAS disk enclosure 69:4 U 24-slot 3.5-inch SAS disk enclosure 96:3 U 2-controller controller enclosure 97:6 U 4-controller controller enclosure 98:2 U 25 Slot 2.5 SSD Disks Enclosure 99:2 U 25-slot 2.5-inch active NVMe controller enclosure 101:2 U 25-slot 2.5-inch SSD NVMe disk enclosure 112:4 U 4-controller controller enclosure 113:2 U 2-controller 25-slot 2.5-inch SAS controller enclosure 114:2 U 2-controller 12-slot 3.5-inch SAS controller enclosure 115:2 U 2-controller 36-slot NVMe controller enclosure 116:2 U 2-controller 25-slot 2.5-inch SAS controller enclosure 117:2 U 2-controller 12-slot 3.5-inch SAS controller enclosure 118:2 U 25-slot 2.5-inch smart SAS disk enclosure 119:2 U 12-slot 3.5-inch smart SAS disk enclosure 120:2 U 36-slot smart NVMe disk enclosure		

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Node	Data Type	Meaning	Permission	Status
		122:2U 2-controller 25-slot 2.5-inch NVMe controller enclosure 130:2 U 2-controller 25-slot 2.5-inch SAS controller enclosure 131:2U 2-controller 12-slot 3.5-inch SAS controller enclosure		
hwInfoEnclosureTemperature OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.8	Integer	Enclosure temperature (unit: ºC)	Read-only	Current
hwInfoEnclosureSN OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.9	OCTET STRING	Enclosure SN	Read-only	Current
hwInfoEnclosureMAC OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.10	OCTET STRING	MAC address of an enclosure	Read-only	Current
hwInfoEnclosureHeight OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.11	Gauge	Enclosure height (unit: U)	Read-only	Current
hwInfoEnclosureExpansionDepth OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.12	Gauge	Number of cascaded enclosures	Read-only	Current
hwInfoEnclosureElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.6.1.13	OCTET STRING	Electronic label of an enclosure	Read-only	Current

hwInfoPortComTable

This table is used to check the serial port information about a device. The index to this table is **hwInfoPortComID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.7.

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Node	Data Type	Description	Permission	Status
hwInfoPortComID OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.1	OCTET STRING	ID of a serial port, table index	Read-only	Current
hwInfoPortComLocation OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.2	OCTET STRING	Location of a serial port	Read-only	Current
hwInfoPortComHealth-Status OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.3	Gauge	Health status of a serial port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortComRunning-Status OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.4	Gauge	Operating status of a serial port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortComType OID:1.3.6.1.4.1.34774.4.1.23.5.7.1.5	Gauge	Logical type of a serial port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current

hwInfoPortEthTable

This table is used to check the Ethernet port information about a device. The index to this table is **hwInfoPortEthID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.8. When the value is empty or "--", "4294967295" or "F", indicates unknown or invalid.

Node	Data Type	Description	Permission	Status
hwInfoPortEthID OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.1	OCTET STRING	ID of an Ethernet port, table index	Read-only	Current
hwInfoPortEthLocation OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.2	OCTET STRING	Location of an Ethernet port	Read-only	Current
hwInfoPortEthHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.3	Gauge	Health status of an Ethernet port. For details, see **HEALTH_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortEthRunning-Status OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.4	Gauge	Operating status of an Ethernet port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortEthType OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.5	Gauge	Logical type of an Ethernet port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current
hwInfoPortEthIPv4Address OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.6	OCTET STRING	IPv4 address of an Ethernet port	Read-only	Current
hwInfoPortEthSubnet-Mask OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.7	OCTET STRING	IPv4 mask of an Ethernet port	Read-only	Current
hwInfoPortEthIPv4Gateway OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.8	OCTET STRING	IPv4 gateway of an Ethernet port	Read-only	Current
hwInfoPortEthIPv6Address OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.9	OCTET STRING	IPv6 address of an Ethernet port	Read-only	Current
hwInfoPortEthIPv6PrefixLength OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.10	OCTET STRING	IPv6 mask of an Ethernet port	Read-only	Current
hwInfoPortEthIPv6Gateway OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.11	OCTET STRING	IPv6 gateway of an Ethernet port	Read-only	Current
hwInfoPortEthMAC OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.12	OCTET STRING	MAC address of an Ethernet port	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortEthRole OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.13	Gauge	Role of an Ethernet port. Possible values are: 2: INI 3: TGT 4: INI_AND_TGT 4294967295: Invalid	Read-only	Current
hwInfoPortEthMode OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.14	Gauge	Auto-negotiation mode of an Ethernet port. Possible values are: 1: HALF_DUPLEX 2: DUPLEX 3: AUTO If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortEthMTU OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.15	Gauge	Maximum transmission unit (MTU) of an Ethernet port (unit: B). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortEthWorkingRate OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.16	Gauge	Working rate of an Ethernet port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortEthBondName OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.17	OCTET STRING	Bond name of an Ethernet port	Read-only	Current
hwInfoPortEthiSCSIPort OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.18	OCTET STRING	iSCSI port number of an Ethernet port. The port is occupied when iSCSI is enabled. The default port number is 3260.	Read-only	Current
hwInfoPortEthiSCSIName OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.19	OCTET STRING	iSCSI name of an Ethernet port	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortEthErrorPackets OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.20	Gauge	Number of error packets of an Ethernet port	Read-only	Current
hwInfoPortEthLostPackages OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.21	Gauge	Number of lost packets of an Ethernet port	Read-only	Current
hwInfoPortEthStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.8.1.22	OCTET STRING	Time when the bit error statistics of an Ethernet port starts. If "--" is returned, this parameter is invalid.	Read-only	Current

hwInfoPortFCTable

This table is used to check the Fibre Channel port information about a device. The index to this table is **hwInfoPortFCID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.9.

Node	Data Type	Description	Permission	Status
hwInfoPortFCID OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.1	OCTET STRING	ID of a Fibre Channel port, table index	Read-only	Current
hwInfoPortFCLocation OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.2	OCTET STRING	Location of a Fibre Channel port	Read-only	Current
hwInfoPortFCHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.3	Gauge	Health status of a Fibre Channel port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortFCRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.4	Gauge	Operating status of a Fibre Channel port. For details, see **RUNNING_STATUS_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortFCType OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.5	Gauge	Logical type of a Fibre Channel port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current
hwInfoPortFCWorkingRate OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.6	Gauge	Working rate of a Fibre Channel port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortFCConfiguredSpeed OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.7	Gauge	Configured rate of a Fibre Channel port (unit: Mbit/s). **0** indicates auto-negotiation. **4294967295** indicates invalid.	Read-only	Current
hwInfoPortFCWWN OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.8	OCTET STRING	World wide name (WWN) of a Fibre Channel port	Read-only	Current
hwInfoPortFCRole OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.9	Gauge	Role of a Fibre Channel port. Possible values are: 2: INI 3: TGT 4: INI_AND_TGT 4294967295: Invalid	Read-only	Current
hwInfoPortFCSFPStatus OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.10	Gauge	Status of a Fibre Channel port. Possible values are: 0: Not Exist 1: Offline 2: Online 4294967295: Invalid	Read-only	Current
hwInfoPortFCWorkingMode OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.11	Gauge	Operating mode of a Fibre Channel port. Possible values are: 0: Fabric 1: FC-AL 2: P2P 3: Auto 4294967295: Invalid	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortFCConfiguredMode OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.12	Gauge	Configured mode of a Fibre Channel port. Possible values are: 0: Fabric 1: FC-AL 2: P2P 3: Auto 4294967295: Invalid	Read-only	Current
hwInfoPortFCFloginDelayTimes OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.13	Gauge	Flogin delay of a Fibre Channel port (unit: ms)	Read-only	Current
hwInfoPortFCLostSignals OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.14	Gauge	Number of Loss of Signal errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCLinkErrorsCodes OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.15	Gauge	Number of Bad RX Char errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCLostSynchronizations OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.16	Gauge	Number of Loss of Sync errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCFailedConnections OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.17	Gauge	Number of Link Fail errors of a Fibre Channel port	Read-only	Current
hwInfoPortFCStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.9.1.18	OCTET STRING	Time when the bit error statistics of a Fibre Channel port starts. If "--" is returned, this parameter is invalid.	Read-only	Current

hwInfoPortPCIETable

This table is used to check the PCIe port information about a device. The index to this table is **hwInfoPortPCIEID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.11. The value of performance statistics entry is "4294967295" or "F", indicates an unknown or invalid.

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Node	Data Type	Description	Permission	Status
hwInfoPortPCIEID OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.1	OCTET STRING	ID of a PCIe port, table index	Read-only	Current
hwInfoPortPCIELocation OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.2	OCTET STRING	Location of a PCIe port	Read-only	Current
hwInfoPortPCIEHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.3	Gauge	Health status of a PCIe port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortPCIERunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.4	Gauge	Operating status of a PCIe port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortPCIESpeed OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.5	Gauge	Data transmission rate of a PCIe port (unit: Mbit/s)	Read-only	Current
hwInfoPortPCIECurrentPeerPortID OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.6	OCTET STRING	ID of the port connecting to a PCIe port	Read-only	Current
hwInfoPortPCIESuggestPeerPortID OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.7	OCTET STRING	ID of the port that is supposed to connect to a PCIe port	Read-only	Current
hwInfoPortPCIELostSignals OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.8	Gauge	Number of bit errors of a PCIe port	Read-only	Current
hwInfoPortPCIEECRCError OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.9	Gauge	Number of ECRC errors of a PCIe port	Read-only	Current
hwInfoPortPCIEBadTlp OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.10	Gauge	Number of bad TLPs of a PCIe port	Read-only	Current
hwInfoPortPCIEBadDllp OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.11	Gauge	Number of bad DLLPs of a PCIe port	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortPCIERecvError OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.12	Gauge	Number of receiver errors of a PCIe port	Read-only	Current
hwInfoPortPCIEChipEccError OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.13	Gauge	Number of chip ECC errors of a PCIe port	Read-only	Current
hwInfoPortPCIEStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.11.1.14	OCTET STRING	Time when the bit error statistics of a PCIe port starts. If "--" is returned, this parameter is invalid.	Read-only	Current

hwInfoPortSASTable

This table is used to check the SAS port information about a device. The index to this table is **hwInfoPortSASID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.12. The value of performance statistics entry is "4294967295" or "F", indicates an unknown or invalid.

Node	Data Type	Description	Permission	Status
hwInfoPortSASID OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.1	OCTET STRING	ID of a SAS port, table index	Read-only	Current
hwInfoPortSASLocation OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.2	OCTET STRING	Location of a SAS port	Read-only	Current
hwInfoPortSASHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.3	Gauge	Health status of a SAS port. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoPortSASRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.4	Gauge	Operating status of a SAS port. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoPortSASType OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.5	Gauge	Logical type of a SAS port. For details, see **PORT_LOGIC_TYPE_E**.	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoPortSASWorkingRate OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.6	Gauge	Working rate of a SAS port (unit: Mbit/s). If the parameter value is **4294967295**, this parameter is invalid.	Read-only	Current
hwInfoPortSASWWN OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.7	OCTET STRING	WWN of a SAS port	Read-only	Current
hwInfoPortSASRole OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.8	Gauge	Role of a SAS port. Possible values are: 2: INI 3: TGT 4: INI_AND_TGT 4294967295: Invalid	Read-only	Current
hwInfoPortSASInvalidDword OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.9	Gauge	Number of invalid DWORDs of a SAS port	Read-only	Current
hwInfoPortSASConsistErrors OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.10	Gauge	Number of consistency errors of a SAS port	Read-only	Current
hwInfoPortSASLossOfDword OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.11	Gauge	Number of lost DWORDs of a SAS port	Read-only	Current
hwInfoPortSASPHYResetErrors OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.12	Gauge	Number of physical reset errors of a SAS port	Read-only	Current
hwInfoPortSASStartTime OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.13	OCTET STRING	Time when the bit error statistics of a SAS port starts. If "--" is returned, this parameter is invalid.	Read-only	Current
hwInfoPortSASEnabled OID:1.3.6.1.4.1.34774.4.1.23.5.12.1.14	Gauge	To determine whether a SAS port is enabled. Possible values are: 0: No 1: Yes	Read-only	Current

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hwInfoInterfaceModuleTable

This table is used to check the interface module information about a device. The index to this table is **hwInfoInterfaceModuleID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.13.

Node	Data Type	Description	Permission	Status
hwInfoInterfaceModuleID OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.1	OCTET STRING	ID of an interface module, table index	Read-only	Current
hwInfoInterfaceModuleLocation OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.2	OCTET STRING	Location of an interface module	Read-only	Current
hwInfoInterfaceModuleHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.3	Gauge	Health status of an interface module. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoInterfaceModuleRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.4	Gauge	Operating status of an interface module. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoInterfaceModuleModel OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.5	Gauge	Type of an interface module. For details, see **INTF_MODEL_E**.	Read-only	Current
hwInfoInterfaceModuleLogicVersion OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.6	OCTET STRING	Logical version of an interface module, When the value is "--", indicates unknown.	Read-only	Current
hwInfoInterfaceModulePCBVersion OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.7	OCTET STRING	PCB version of an interface module	Read-only	Current
hwInfoInterfaceModuleElectronicLabel OID:1.3.6.1.4.1.34774.4.1.23.5.13.1.8	OCTET STRING	Electronic label of an interface module	Read-only	Current

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hwInfoExpBoardTable

This table is used to check the expansion module information about a device. The index to this table is **hwInfoExpBoardID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.5.14.

Node	Data Type	Description	Permission	Status
hwInfoExpBoardID OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.1	OCTET STRING	ID of an expansion module, table index	Read-only	Current
hwInfoExpBoardLocation OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.2	OCTET STRING	Location of an expansion module	Read-only	Current
hwInfoExpBoardHealthStatus OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.3	Gauge	Health status of an expansion module. For details, see **HEALTH_STATUS_E**.	Read-only	Current
hwInfoExpBoardRunningStatus OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.4	Gauge	Operating status of an expansion module. For details, see **RUNNING_STATUS_E**.	Read-only	Current
hwInfoExpBoardModel OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.5	Gauge	Model of an expansion module. Possible values are: ● 0: SAS ● 1: FC ● 5: NVMe ● 6: Smart SAS ● 7: Smart NVMe	Read-only	Current
hwInfoExpBoardLogicVersion OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.6	OCTET STRING	Logical version of an expansion module, When the value is "--", indicates unknown	Read-only	Current
hwInfoExpBoardPCBVersion OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.7	OCTET STRING	PCB version of an expansion module	Read-only	Current
hwInfoExpBoardSESVersion OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.8	OCTET STRING	SES version of an expansion module	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoExpBoardELabel OID:1.3.6.1.4.1.34774.4.1.23.5.14.1.9	OCTET STRING	Electronic label of an expansion module	Read-only	Current

5.8 NAS Information Query Module

hwInfoShareNFSTable

This table is used to check the NFS share information about a device. The index to this table is **hwInfoShareNFSShareID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.1.

Node	Data Type	Description	Permission	Status
hwInfoShareNFSShareID OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.1	OCTET STRING	Share ID, table index	Read-only	Current
hwInfoShareNFSFileSystemID OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.2	OCTET STRING	File system ID	Read-only	Current
hwInfoShareNFSDescription OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.3	OCTET STRING	Description	Read-only	Current
hwInfoShareNFSLocalPath OID: 1.3.6.1.4.1.34774.4.1.23.7.1.1.4	OCTET STRING	Share path	Read-only	Current

hwInfoShareCIFSTable

This table is used to check the CIFS share information about a device. The index to this table is **hwInfoShareCIFSID**.

The the OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.2.

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Node	Data Type	Description	Permission	Status
hwInfoShareCIFSID OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.1	OCTET STRING	Share ID, table index	Read-only	Current
hwInfoShareCIFSName OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.2	OCTET STRING	Object name	Read-only	Current
hwInfoShareCIFSFileSystemID OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.3	OCTET STRING	File system ID	Read-only	Current
hwInfoShareCIFSDescription OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.4	OCTET STRING	Description	Read-only	Current
hwInfoShareCIFSLocalPath OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.5	OCTET STRING	Share path	Read-only	Current
hwInfoShareCIFSOplockEnabled OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.6	Gauge	Optimization lock switch • 0: No • 1: Yes	Read-only	Current
hwInfoShareCIFSNotifyEnabled OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.7	Gauge	Notification switch • 0: No • 1: Yes	Read-only	Current
hwInfoShareCIFSContAvailableEna OID: 1.3.6.1.4.1.34774.4.1.23.7.2.1.8	Gauge	Whether failover is enabled • 0: No • 1: Yes	Read-only	Current

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hwInfoSharePermsNFSTable

This table is used to check the NFS share permission of a device. The index to this table is **hwInfoSharePermsNFSID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.3.

Node	Data Type	Description	Permission	Status
hwInfoSharePermsNFSID OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoSharePermsNFSAccessName OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.2	OCTET STRING	Access name	Read-only	Current
hwInfoSharePermsNFSShareID OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.3	OCTET STRING	Share ID	Read-only	Current
hwInfoSharePermsNFSAccessType OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.4	Gauge	Access type • 0: read-only • 1: read and write • 4: read and write but cannot be deleted or renamed, supported by massive storage only)	Read-only	Current
hwInfoSharePermsNFSSyncEnabled OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.5	Gauge	Synchronization switch • 0: Data is synchronized • 1: Data is not synchronized	Read-only	Current
hwInfoSharePermsNFSAllSquashEna OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.6	Gauge	Whether the GID and UID of the shared directory are mapped as anonymous users • 0: yes • 1: no	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoSharePermsNFS-RootSquashEna OID: 1.3.6.1.4.1.34774.4.1.23.7.3.1.7	Gauge	Permission squash of user root • 0: rootSquash • 1: no rootSquash	Read-only	Current

hwInfoSharePermsCIFSTable

This table is used to check the CIFS share permission of a device. The index to this table is **hwInfoSharePermsCIFSID**.

The prefix to the OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.4. Obtain the actual OID of each node in the MIB file.

Node	Data Type	Description	Permission	Status
hwInfoSharePermsCIFSID OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoSharePermsCIFSAccessName OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.2	OCTET STRING	Name of a user or user group	Read-only	Current
hwInfoSharePermsCIFS-ShareID OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.3	OCTET STRING	Share ID	Read-only	Current
hwInfoSharePermsCIFS-DomainType OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.4	Gauge	Domain type • 0: AD • 1: LDAP • 2: local user	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoSharePermsCIFSPermsType OID: 1.3.6.1.4.1.34774.4.1.23.7.4.1.5	Gauge	Permission type • 0: read • 1: full control • 2: forbidden • 3: default • 4: read and write (cannot be deleted, renamed) • 5: read and write, being used in unified storage but not massive storage	Read-only	Current

hwInfoLogicalPortTable

This table is used to check the logical port information about a device. The index to this table is **hwInfoLogicalPortID**.

The the OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.5.

Node	Data Type	Description	Permission	Status
hwInfoLogicalPortID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoLogicalPorttName OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.2	OCTET STRING	Object name	Read-only	Current
hwInfoLogicalPortRunStatus OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.3	Gauge	Operating status (for details, see **RUNNING_STATUS_E**)	Read-only	Current
hwInfoLogicalPortIPv4Addr OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.4	OCTET STRING	IPv4 address	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortIPv4Mask OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.5	OCTET STRING	IPv4 mask	Read-only	Current
hwInfoLogicalPortIPv4Gateway OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.6	OCTET STRING	IPv4 gateway	Read-only	Current
hwInfoLogicalPortIPv6Addr OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.7	OCTET STRING	IPv6 address	Read-only	Current
hwInfoLogicalPortIPv6Mask OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.8	OCTET STRING	IPv6 mask	Read-only	Current
hwInfoLogicalPortIPv6Gateway OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.9	OCTET STRING	IPv6 gateway	Read-only	Current
hwInfoLogicalPortRole OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.10	Gauge	Role of the logical port • 0: unknown • 1: management • 2: service • 3: all	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortSupportProt OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.11	Gauge	Protocol: • 0: none • 1: NFS • 2: CIFS • 3: NFS + CIFS • 4: iSCSI • 8: FC/FCoE • 16: HTTPS • 32: NDMP • 35: NFS+CIFS+NDMP • 51: HTTPS+NFS+CIFS+NDMP • 64: NVMe	Read-only	Current
hwInfoLogicalPortHomePortType OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.12	Gauge	Type of the parent port: • 1: ETH • 2: FC • 3: FCoE • 4: SAS • 5: PCIe • 6: IB • 7: bond • 8: VLAN • 9: LIF	Read-only	Current
hwInfoLogicalPortHomePortID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.13	OCTET STRING	ID of the parent port	Read-only	Current
hwInfoLogicalPortOwnerCtrlID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.14	OCTET STRING	ID of the parent node	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortCurrPortType OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.15	Gauge	Port type • 1: ETH • 2: FC • 3: FCoE • 4: SAS • 5: PCIe • 6: IB • 7: bond • 8: VLAN • 9: LIF	Read-only	Current
hwInfoLogicalPortCurrPortID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.16	OCTET STRING	Port ID	Read-only	Current
hwInfoLogicalPortWorkCtrlID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.17	OCTET STRING	Node ID	Read-only	Current
hwInfoLogicalPortActState OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.18	Gauge	Active or not • 0: No • 1: Yes	Read-only	Current
hwInfoLogicalPortAddrFamily OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.19	Gauge	Address type • 0: IPv4 • 1: IPv6	Read-only	Current
hwInfoLogicalPortIsPrivate OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.20	Gauge	Configurable or not • 0: No • 1: Yes	Read-only	Current
hwInfoLogicalPortFailOVGID OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.21	OCTET STRING	ID of the owning failover group	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoLogicalPortFailOVEnable OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.22	Gauge	Whether failover is enabled • 0: No • 1: Yes	Read-only	Current
hwInfoLogicalPortFailBackMode OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.23	Gauge	Failover mode: • 1: manual • 2: automatic	Read-only	Current
hwInfoLogicalPortFailOVGName OID: 1.3.6.1.4.1.34774.4.1.23.7.5.1.24	OCTET STRING	Name of the owning failover group	Read-only	Current

hwInfoFileSysTable

This table is used to check the file system information about a device. The index to this table is **hwInfoFileSysID**.

The OID of this table is 1.3.6.1.4.1.34774.4.1.23.7.6.

Node	Data Type	Description	Permission	Status
hwInfoFileSysID OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.1	OCTET STRING	Object ID, table index	Read-only	Current
hwInfoFileSysName OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.2	OCTET STRING	File system name	Read-only	Current
hwInfoFileSysHeathStatus OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.3	Gauge	Health status (for details, see **HEALTH_STATUS_E**)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoFileSysRunningStatus OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.4	Gauge	Operating status (for details, see **RUNNING_STATUS_E**)	Read-only	Current
hwInfoFileSysDescription OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.5	OCTET STRING	Description	Read-only	Current
hwInfoFileSysSubType OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.6	Gauge	Subtype: • 0: Normal • 1: WORM	Read-only	Current
hwInfoFileSysAllocType OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.7	Gauge	Space allocation type: • 0: thick • 1: thin	Read-only	Current
hwInfoFileSysCapacity OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.8	Counter64	Configured capacity (unit: MB)	Read-only	Current
hwInfoFileSysSnapshotReservePer OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.9	Gauge	Space reserved for snapshot (unit: %)	Read-only	Current
hwInfoFileSysSnapshotUseCapacity OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.10	Counter64	Capacity used by snapshots (unit: MB)	Read-only	Current
hwInfoFileSysSectorSize OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.11	Gauge	Block size (unit: byte)	Read-only	Current
hwInfoFileSysOwningContrller OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.12	OCTET STRING	Owning controller	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoFileSysWorking-Controller OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.13	OCTET STRING	Working controller	Read-only	Current
hwInfoFileSysIOPriotiry OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.14	Gauge	File system I/O priority: • 1: low • 2: medium • 3: high	Read-only	Current
hwInfoFileSysEnableCompression OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.15	Gauge	Whether compression is enabled • 0: No • 1: Yes	Read-only	Current
hwInfoFileSysCompression OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.16	Gauge	Compression algorithm: • 0: fast • 1: slow	Read-only	Current
hwInfoFileSysIsShowSnapDir OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.17	Gauge	Snapshot directory visible or not • 0: No • 1: Yes	Read-only	Current
hwInfoFileSysAvailableCapacity OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.18	Counter64	Available file system capacity (unit: MB)	Read-only	Current
hwInfoFileSysAvAndAllcCapRatio OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.19	Gauge	Percentage of available file system used capacity to initial capacity	Read-only	Current
hwInfoFileSysSCCachedSize OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.20	Counter64	SmartCache size (unit: KB)	Read-only	Current

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Node	Data Type	Description	Permission	Status
hwInfoFileSysSCHitRage OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.21	Gauge	SmartCache hit ratio (unit: %)	Read-only	Current
hwInfoFileSysCompressSavedRatio OID: 1.3.6.1.4.1.34774.4.1.23.7.6.1.22	Gauge	Compression ratio (unit: %)	Read-only	Current

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6 Appendix

6.1 Common Enumerations

6.1 Common Enumerations

6.1.1 Definition of Enum Values (HEALTH_STATUS_E)

Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	--	7	Bit Errors Found	14	Invalid
1	Normal	8	Consistent	15	Write Protect
2	Fault	9	Inconsistent	16	single link
3	Pre-Fail	10	Busy	17	No Redundant Link
4	Partially Broken	11	No Input	18	Offline
5	Degraded	12	Low Battery		
6	Bad Sectors Found	13	Single Link Fault		

6.1.2 Definition of Enum Values (RUNNING_STATUS_E)

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	--	48	Charging	96	Partition Migrating
1	Normal	49	Charging completed	97	Mount
2	Running	50	Discharging	98	Umount
3	Not running	51	Upgrading	99	INSTALLING
4	Not existed	52	Power Lost	100	To Be Synchronized
5	Sleep in high temperature	53	Initializing	101	Connecting
6	Starting	54	Apply change	102	Service Switching
7	Power failure rotection	55	Online disable	103	Power-on failed
8	Spin down	56	Offline disable	104	REPAIRING
9	Started	57	Online frozen	105	abnormal
10	Link Up	58	Offline frozen	106	Deleting
11	Link Down	59	Closed	107	Modifying
12	Powering on	60	Removing	108	Running(clearing data)
13	Powered off	61	In service	109	Running(synchronizing data)
14	Pre-copy	62	Out of service	110	Standby
15	Copyback	63	Running normal	111	STOPPING
16	Reconstruction	64	Running fail	112	RESTORE FAULT
17	Expansion	65	Running success	113	Cut Over
18	Unformatted	66	Running success	114	Erasing
19	Formatting	67	Running failed	115	Verifying
20	Unmapped	68	Waiting	119	Creating
21	Initial synchronizing	69	Canceling		
22	Consistent	70	Canceled		
23	Synchronizing	71	About to synchronize		

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
24	Synchronized	72	Synchronizing data		
25	Unsynchronized	73	Failed to synchronize		
26	Split	74	Fault		
27	Online	75	Migrating		
28	Offline	76	Migrated		
29	Locked	77	Activating		
30	Enabled	78	Deactivating		
31	Disabled	79	Start failed		
32	Balancing	80	Stop failed		
33	To be recovered	81	Decommissioning		
34	Interrupted	82	Decommissioned		
35	Invalid	83	Recommissioning		
36	Not start	84	Replacing node		
37	Queuing	85	Scheduling		
38	Stopped	86	Pausing		
39	Copying	87	Suspending		
40	Completed	88	Suspended		
41	Paused	89	Overload		
42	Reverse synchronizing	90	To be switch		
43	Activated	91	Switching		
44	Restore	92	To be cleanup		
45	Inactive	93	Forced start		
46	Idle	94	Error		
47	Powering off	95	Job completed		

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6.1.3 Definition of Enum Values (PORT_LOGIC_TYPE_E)

Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	Host Port	6	Maintenance/ Host Port		
1	Expansion Port	7	Backup MGR Port		
2	Management Port	8	Product Storage Port		
3	Inner Port	9	Backup Storage Port		
4	Maintenance Port	10	ETH NOT Config Port		
5	Management/ Host Port	11	IP Scale Out		

6.1.4 Definition of Enum Values (INTF_MODEL_E)

Enum Values	English Term	Enum Values	English Term
1	4 x 4 G FC Interface Module	2339	4 ports FE 10 Gbit/s ROCE I/O module
2	2 x 4 G FC Interface Module	2340	4 ports FE 25 Gbit/s ROCE I/O module
3	2 x GE Electrical Interface Module	2341	4 ports FE 8 Gbit/s Fibre Channel I/O module
4	4 x SAS I Interface Module	2342	4 ports FE 16 Gbit/s Fibre Channel I/O module
5	Swapped FC Interface Module	2343	4 ports FE 32 Gbit/s Fibre Channel I/O module
6	2 x 10GE Optical Interface Module	2344	4 ports FE 10 Gbit/s ETH I/O module
7	1 x 8 G FC Optical Interface Module	2345	4 ports FE 25 Gbit/s ETH I/O module
8	2 x 8 G FC Optical Interface Module	2346	4 ports FE 10 Gbit/s ROCE I/O module

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Enum Values	English Term	Enum Values	English Term
9	2 x SAS II Electrical Interface Module	2347	4 ports FE 25 Gbit/s ROCE I/O module
10	4 x SAS II Electrical Interface Module	2348	2 ports FE 40 Gbit/s ETH I/O module
11	2 x FC + 2 x GE Interface Module	2349	2 ports FE 100 Gbit/s ETH I/O module
12	4 x GE Electrical Interface Module	2350	2 ports BE 100 Gbit/s RDMA I/O module
13	4 x 8 G FC Optical Interface Module	2351	2 ports SO 100 Gbit/s RDMA I/O module
16	4 x mini-SAS Interface Module	2352	2 ports FE 40 Gbit/s ROCE I/O module
21	4 x 10 G FCoE Optical Interface Module	2353	2 ports FE 100 Gbit/s ROCE I/O module
22	2 x 6 G SAS Interface Module	2354	2 ports FE 40 Gbit/s ETH I/O module
23	2 x 6 G SAS Interface Module	2355	2 ports FE 100 Gbit/s ETH I/O module
24	Management Board	2356	2 ports BE 100 Gbit/s RDMA I/O module
25	4 x 10GE Interface Module	2357	2 ports SO 100 Gbit/s RDMA I/O module
26	PCIe Interface Module	2358	2 ports FE 40 Gbit/s ROCE I/O module
27	2x10G Optical Interface Module	2359	2 ports FE 100 Gbit/s ROCE I/O module
28	4xGE Optical Interface Module	2360	4 ports FE 10 Gbit/s ETH I/O module
29	2 x 16 G FC Optical Interface Module	2361	4 ports SO 25 Gbit/s RDMA I/O module
30	4 x 12SAS QSFP Interface Module	2362	2 ports SO 100 Gbit/s RDMA I/O module
31	4 x 10GE Electrical Interface Module	2363	2 ports SO 100 Gbit/s RDMA I/O module
32	2 x 40GE Infiniband Interface Module	2364	4 ports FE 10 Gbit/s ETH I/O module

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Enum Values	English Term	Enum Values	English Term
33	2 port 4*14Gb IB I/O module	2365	4 ports FE 25 Gbit/s ETH I/O module
34	4 x GE Electrical Interface Module	2366	4 ports FE 10Gbit/s ETH I/O module
35	Smart ACC module	2367	4 ports FE 25 Gbit/s ROCE I/O module
36	4x10GE Electrical Interface Module	2368	4 ports FE 8 Gbit/s Fibre Channel I/O module
37	4 port SmartIO I/O module	2369	4 ports FE 16 Gbit/s Fibre Channel I/O module
38	8x8G FC Optical Interface Module	2370	4 ports FE 32 Gbit/s Fibre Channel I/O module
39	4x8G FC Optical Interface Module	2371	4 ports FE 10 Gbit/s ETH I/O module
40	4x16G FC Optical Interface Module	2372	4 ports FE 25 Gbit/s ETH I/O module
41	12 port 4*12Gb SAS Back-End Interconnect I/O Module	2373	4 ports FE 25 Gbit/s ROCE I/O module
42	12 port SmartIO I/O module	2375	4 ports FE 8 Gbit/s Fibre Channel I/O module
43	RoCE 10G/25G Interface Module	2376	4-port FE 16 Gbit/s FC I/O module
44	2-port PCIe 3.0 Interface Module	2377	4-port FE 32 Gbit/s FC I/O module
55	2-port 8 Gbit/s PCIe scale-up interface module	2378	4-port FE 10 Gbit/s ETH I/O module
58	8x16G FC Optical Interface Module	2379	4-port FE 25 Gbit/s ETH I/O module
65	4 ports SmartIO I/O module	2380	4-port FE 10 Gbit/s ROCE I/O module
66	4 ports SmartIO I/O module	2381	4 ports FA 25 Gbit/s ROCE I/O module
67	2 ports 100Gb/40Gb ETH I/O module	2382	2-port FE 40 Gbit/s ETH I/O module
68	2 ports 100Gb/40Gb ETH I/O module	2383	2-port FE 100 Gbit/s ETH I/O module

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Enum Values	English Term	Enum Values	English Term
71	4 ports 10Gb ETH I/O module	2384	2-port BE 100 Gbit/s RDMA I/O module
516	4 ports FE 1 Gbit/s ETH I/O module	2385	2-port SO 100 Gbit/s RDMA I/O module
518	4 ports BE 12 Gbit/s SAS I/O module	2386	2-port FE 40 Gbit/s ROCE I/O module
529	AI Accelerator Card	2387	2-port FE 100 Gbit/s ROCE I/O module
535	AI Accelerator Card	2388	2 ports FA 100 Gbit/s ROCE I/O module
537	4 ports FE 1 Gbit/s ETH I/O module	2389	4 ports BE 12 Gbit/s SAS I/O module
538	4 ports BE 12 Gbit/s SAS I/O module	2390	4 ports FE 1 Gbit/s ETH I/O module
580	4 ports FE 1 Gbit/s ETH I/O module	2391	2 ports FE 1 Gbit/s ETH I/O module
583	4 ports BE 12 Gbit/s SAS V2 I/O module	2392	4-port SO 25 Gbit/s RDMA I/O module
601	4 ports FE 1 Gbit/s ETH I/O module	2393	4 ports FE 25 Gbit/s ROCE I/O module
2304	4 ports FE 8 Gbit/s Fibre Channel I/O module	2394	4 ports FE 8 Gbit/s FC I/O module
2305	4 ports FE 16 Gbit/s Fibre Channel I/O module	2395	4-port FE 16 Gbit/s FC I/O module
2306	4 ports FE 32 Gbit/s Fibre Channel I/O module	2396	4-port FE 32 Gbit/s FC I/O module
2307	4 ports FE 10 Gbit/s ETH I/O module	2397	4-port FE 10 Gbit/s ETH I/O module
2308	4 ports FE 25 Gbit/s ETH I/O module	2398	4-port FE 25 Gbit/s ETH I/O module
2309	4 ports SO 25 Gbit/s RDMA I/O module	2399	4-port SO 25 Gbit/s RDMA I/O module
2310	4 ports FE 8 Gbit/s Fibre Channel I/O module	2400	4 ports FE 10 Gbit/s ROCE I/O module
2311	4 ports FE 16 Gbit/s Fibre Channel I/O module	2401	4 ports FE 25 Gbit/s ROCE I/O module

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Enum Values	English Term	Enum Values	English Term
2312	4 ports FE 32 Gbit/s Fibre Channel I/O module	2402	2-port FE 40 Gbit/s ETH I/O module
2313	4 ports FE 10 Gbit/s ETH I/O module	2403	2-port FE 100 Gbit/s ETH I/O module
2314	4 ports FE 25 Gbit/s ETH I/O module	2404	2-port BE 100 Gbit/s RDMA I/O module
2315	2 ports FE 40 Gbit/s ETH I/O module	2405	2-port SO 100 Gbit/s RDMA I/O module
2316	2 ports FE 100 Gbit/s ETH I/O module	2406	2-port FE 40 Gbit/s ROCE I/O module
2317	2 ports BE 100 Gbit/s RDMA I/O module	2407	2-port FE 100 Gbit/s ROCE I/O module
2318	2 ports SO 100 Gbit/s RDMA I/O module	2408	4-port FE 10 Gbit/s ETH I/O module
2319	2 ports FE 40 Gbit/s ETH I/O module	2409	4-port FE 25 Gbit/s ETH I/O module
2320	2 ports FE 100 Gbit/s ETH I/O module	2410	4-port FE 25 Gbit/s RoCE I/O module
2321	2 ports BE 100 Gbit/s RDMA I/O module	2411	4-port FE 10 Gbit/s ETH I/O module
2322	2 ports SO 100 Gbit/s RDMA I/O module	2412	4-port FE 10 Gbit/s ETH I/O module
2323	4 ports FE 10 Gbit/s ROCE I/O module	2413	4-port FE 25 Gbit/s ETH I/O module
2324	4 ports FE 25 Gbit/s ROCE I/O module	2414	4-port FE 25 Gbit/s RoCE I/O module
2325	4 ports FE 10 Gbit/s ROCE I/O module	4133	System Management Module
2326	4 ports FE 25 Gbit/s ROCE I/O module	4134	System Management Module
2327	2 ports FE 40 Gbit/s ROCE I/O module	4294967295	unknown
2328	2 ports FE 100 Gbit/s ROCE I/O module		

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Enum Values	English Term	Enum Values	English Term
2329	2 ports FE 40 Gbit/s ROCE I/O module		
2330	2 ports FE 100 Gbit/s ROCE I/O module		
2331	4 ports FE 10 Gbit/s ETH I/O module		
2332	4 ports FE 10 Gbit/s ETH I/O module		
2333	4 ports FE 8 Gbit/s Fibre Channel I/O module		
2334	4 ports FE 16 Gbit/s Fibre Channel I/O module		
2335	4 ports FE 32 Gbit/s Fibre Channel I/O module		
2336	4 ports FE 10 Gbit/s ETH I/O module		
2337	4 ports FE 25 Gbit/s ETH I/O module		
2338	4 ports SO 25 Gbit/s RDMA I/O module		

6.1.5 Definition of Enum Values (PRODUCT_MODE_E)

Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
0	V1500	90	18800F V3	828	OceanStor Dorado 5800 V6
1	V1800	91	FusionStorage File	829	OceanStor Dorado 6800 V6
2	S2100	92	2800 V5	830	OceanStor Dorado 18500 V6
3	S2300	93	5300 V5	831	OceanStor Dorado 18800 V6

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
4	S5100	94	5300F V5	832	OceanStor Dorado 18800K V6
5	S5300	95	5500 V5	833	OceanStor 5310
6	S5500	96	5500F V5	834	OceanStor 5510
7	S5600	97	5600 V5	835	OceanStor 5610
8	VIS	98	5600F V5	836	OceanStor 6810
9	S2600	99	5800 V5	837	OceanStor 18510
10	S6800E	100	5800F V5	838	OceanStor 18810
11	V1500N	101	6800 V5	839	OceanStor Dorado 5600K
12	S2300E	102	6800F V5	840	OceanStor 18500K
13	COMMON	103	18500 V5	841	OceanStor 5000 V3
15	N8000	104	18500F V5	842	OceanStor 6000 V3
16	S6900	105	18800 V5	843	OceanStor 5810-HS
17	S3900	106	18800F V5	844	OceanStor 5300K
18	S5500T	107	5500 V5 Elite	845	OceanStor 5500K
19	S5600T	108	2100 V3	846	OceanStor Micro 1300
20	S6800T	109	2288H V5 2U 12	847	OceanStor Micro 1500
21	S6900-M100	110	2288H V5 2U 25	848	OceanStor 2910
22	S3900-M200	111	5288 V5	849	OceanStor 5320
23	S3900-M300	112	2200 V3	850	OceanStor 5510S
24	S5900-M100	113	2600 V3	851	OceanStor Dorado 2000
25	S8100	114	2600F V3	852	OceanProtect X3000
26	S5800T	115	5300 V5	853	OceanStor 2200
27	V1600N	116	5110 V5	854	OceanStor 2220
28	S5900-M200	117	5110F V5	855	OceanStor 2600

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
29	N8001	118	5210 V5	856	OceanStor 5110
30	S2600T	119	5210F V5	857	OceanStor 5120
31	S8000-I	120	5310 V5	858	OceanStor 5210
32	Dorado5100	121	5310F V5	859	OceanStor 5220
33	T3200	122	5510 V5	860	OceanStor Micro 1600
34	T3500	123	5510F V5	861	OceanStor 2620
35	N8000-R710	124	5610 V5	862	OceanDisk 1500T
36	N8000-C2	125	5610F V5	863	OceanDisk 1600T
40	8000-C2-SYMC	126	5810 V5	864	OceanDisk 1300
41	VTL	127	5810F V5	865	OceanDisk 1500
42	S2900	128	6810 V5	880	OceanData 5910
43	S2200T	129	6810F V5	1000	5300 V5
44	Dorado2100_G2	130	18510 V5	1001	5300K V5
45	VIS 6000	131	18510F V5	1002	5300F V5
46	HVS85T	132	18810 V5	1003	5500 V5
47	HVS88T	133	18810F V5	1004	5500K V5
48	9000	134	5210 V5 Enhanced	1005	5500F V5
49	T3200 G2	135	5210F V5 Enhanced	1006	5600 V5
50	T3500 G2	137	2810 V5	1007	5600K V5
51	T3200 G3	138	FusionCube	1008	5600F V5
52	T3500 G3	139	5110 V5 Enhanced	1009	5800 V5
53	S5700	140	5110F V5 Enhanced	1010	5800K V5
54	CE6800	141	2600 V5	1011	5800F V5

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
55	CE12800	142	2810 V5 Enhanced	1012	6800 V5
56	18800F	143	2210	1013	6800K V5
57	18500	150	9500	1014	6800F V5
58	18800	151	9000-K	1015	18500 V5
60	Dorado2100	152	9000-A	1016	18500K V5
61	6800 V3	401	N5000	1017	18500F V5
62	6900 V3	601	FusionStorage OBS	1018	18800 V5
63	5600 V3	800	VIS8200	1019	18800K V5
64	5800 V3	801	VIS8400	1020	18800F V5
65	T3200 G3	802	VIS8600	1021	2800 V5
66	WUSHAN_SQL	803	VIS8800	1022	2200 V5
67	WUSHAN_HD	804	Cloud Protecto	1200	OceanProtect X8000
68	5500 V3	805	Dorado5000 V3	1201	OceanProtect X9000
69	2600 V3	806	Dorado6000 V3	1202	OceanProtect A8000
70	5300 V3	807	Dorado18000 V3	1203	OceanProtect A9000
71	2800 V3	808	Dorado NAS	1204	OceanProtect A6000
72	18500 V3	809	Dorado NAS Enhanced	1205	OceanProtect X6000
73	18800 V3	810	Dorado3000 V3	1206	OceanProtect X8000K
74	High-Density Controller Enclosure	811	OceanStor Dorado 5000 V6	1207	OceanProtect X9000K

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
75	RH2288V3_25	812	OceanStor Dorado 5000 V6	1208	OceanProtect L8000
76	RH2288V3_12	813	OceanStor Dorado 6000 V6	1301	CyberEngine 300
77	RH5288V3_36	814	OceanStor Dorado 6000 V6		
78	2200 V3	815	OceanStor Dorado 8000 V6		
79	5100 V3	816	OceanStor Dorado 8000 V6		
80	UDS	817	OceanStor Dorado 18000 V6		
81	OceanStor DJ	818	OceanStor Dorado 18000 V6		
82	2600 V3 for Video	819	OceanStor Dorado 3000 V6		
83	Fusion Storage	821	OceanStor Dorado 5000 V6		
84	2600F V3	822	OceanStor Dorado 6000 V6		
85	5500F V3	823	OceanStor Dorado 8000 V6		
86	5600F V3	824	OceanStor Dorado 18000 V6		
87	5800F V3	825	OceanStor Dorado 5300 V6		

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Enum Values	English Term	Enum Values	English Term	Enum Values	English Term
88	6800F V3	826	OceanStor Dorado 5500 V6		
89	18500F V3	827	OceanStor Dorado 5600 V6		

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